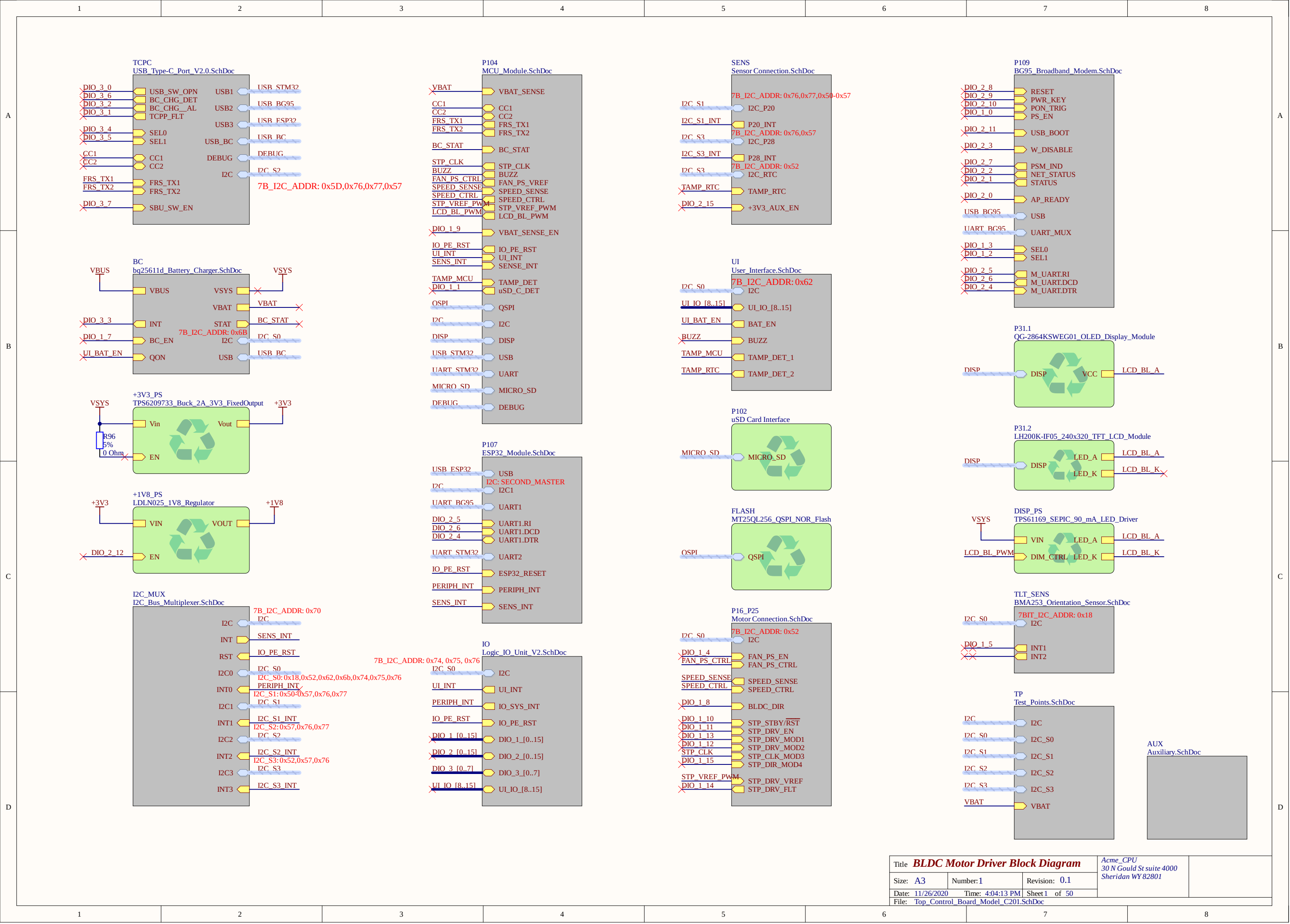




A

B

C

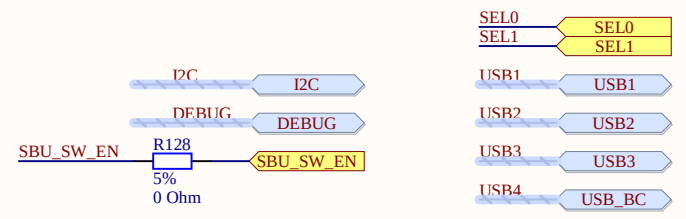
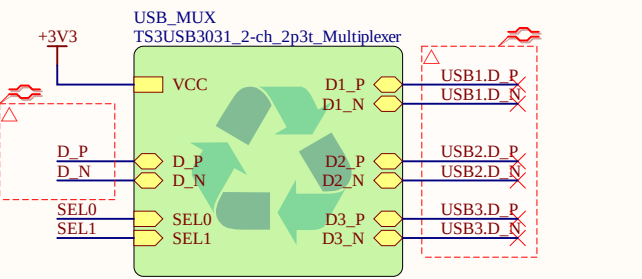
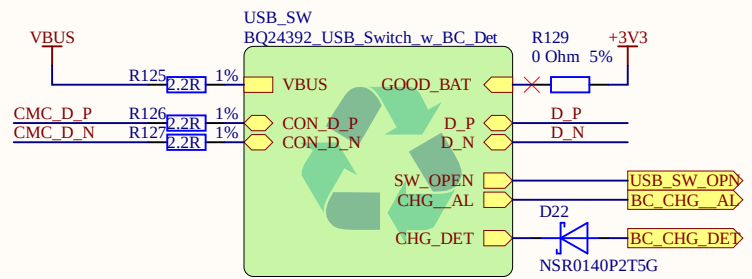
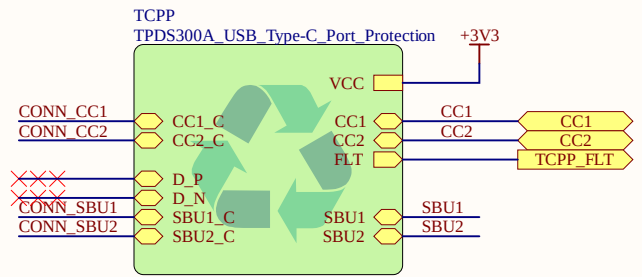
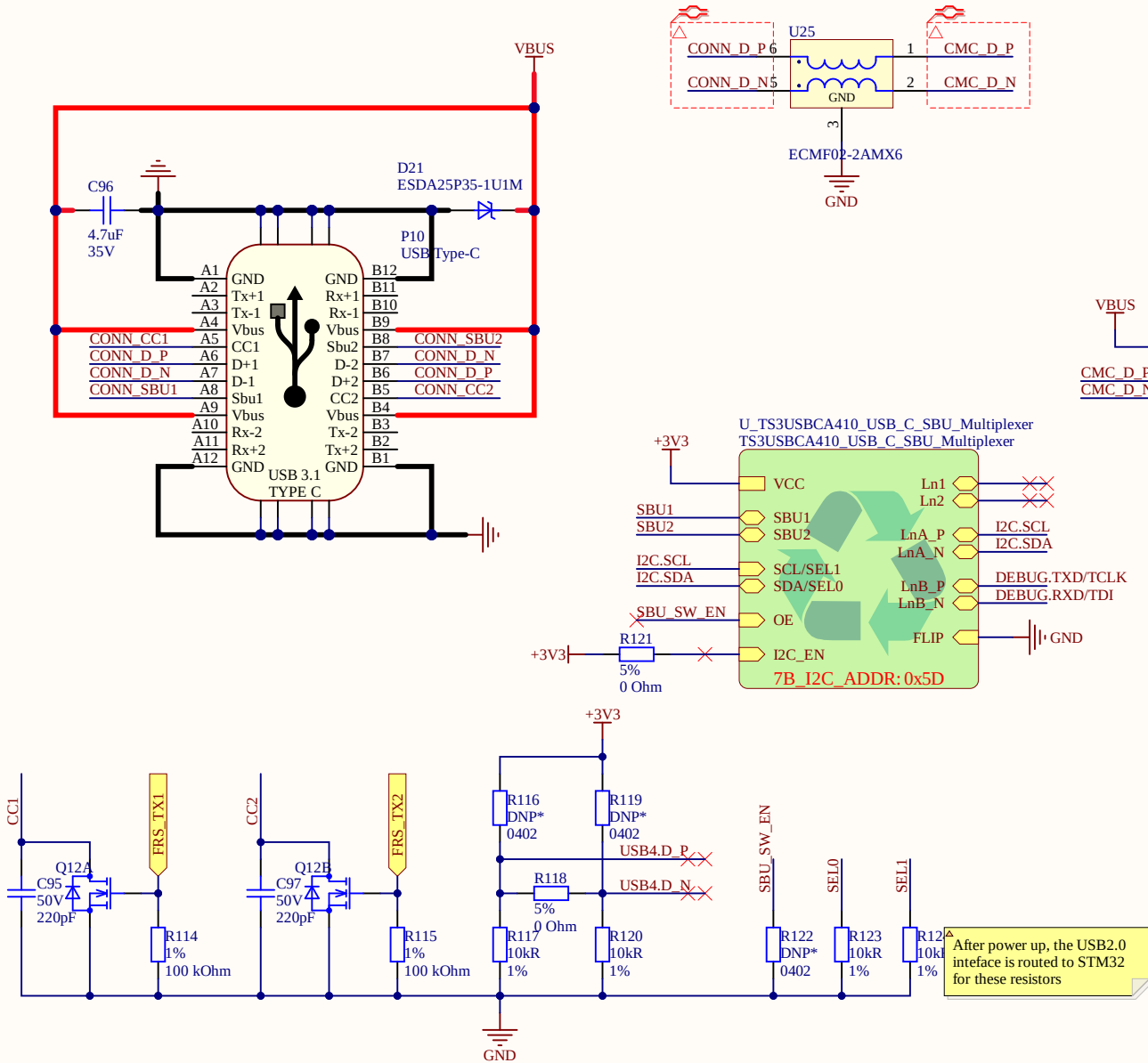
D

A

B

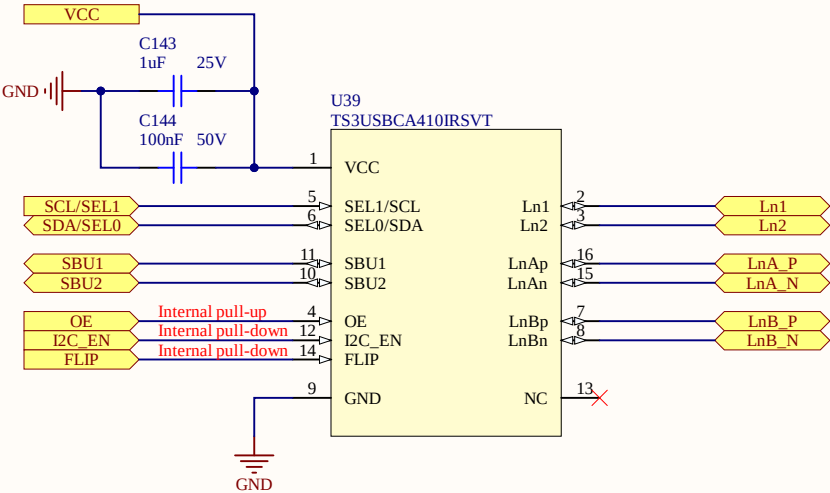
C

D

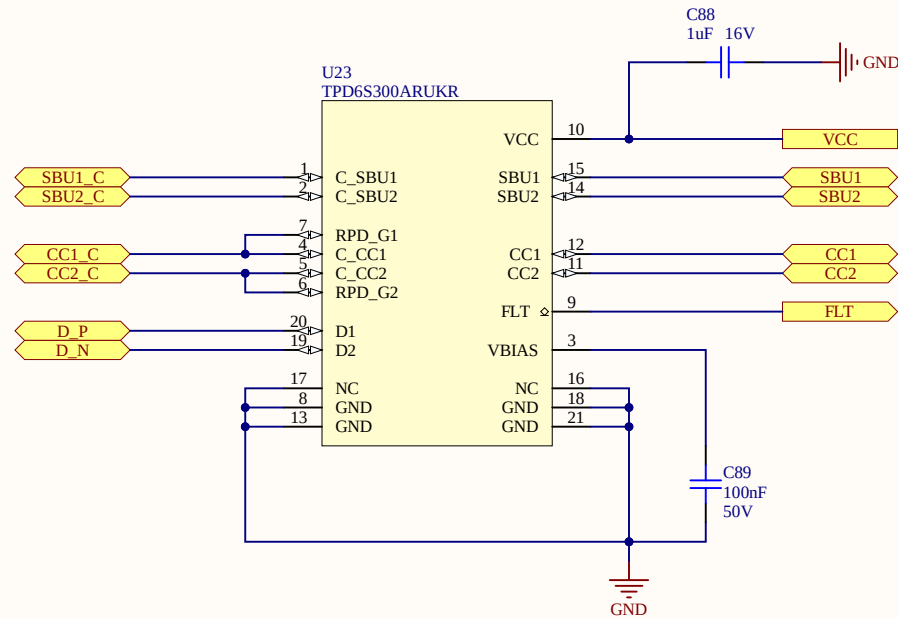


After power up, the USB2.0 interface is routed to STM32 for these resistors

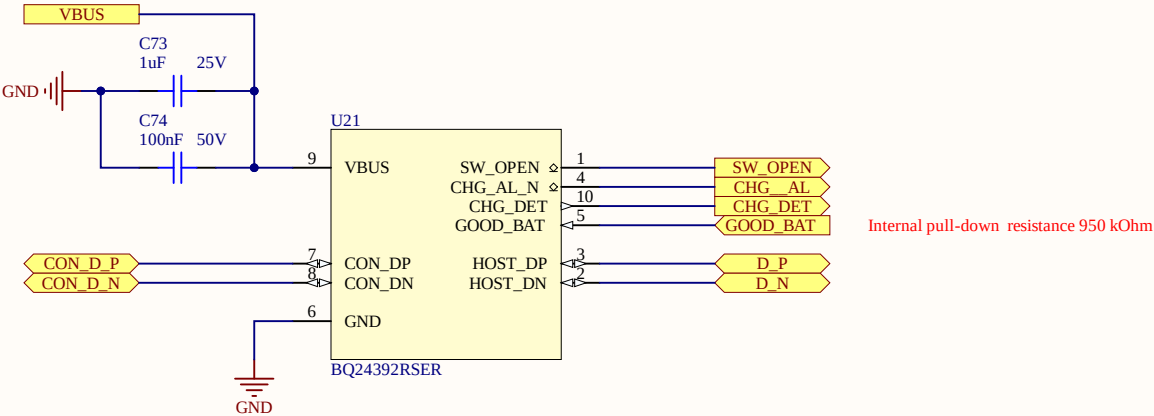
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Size: A4	Number: 2	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:13 PM	Sheet 2 of 50		
File: USB_Type-C_Port_V2.0.SchDoc				



Title BQ24392 SPST USB Switch			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801
Size: A4	Number:*	Revision: 0.1	
Date: 11/26/2020	Time: 4:04:13 PM	Sheet* of *	
File: TS3USBCA410_USB_C_SBU_Multiplexer.SchDoc			



Title TPD6S300A USB Type-C Protection			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number: *	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:13 PM	Sheet * of *		
File: TPDS300A_USB_Type-C_Port_Protection.SchDoc				



Title BQ24392 SPST USB Switch			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801
Size: A4	Number:*	Revision: 0.1	
Date: 11/26/2020	Time: 4:04:13 PM	Sheet* of *	
File: BQ24392_USB_Switch_w_BC_Det.SchDoc			

A

A

B

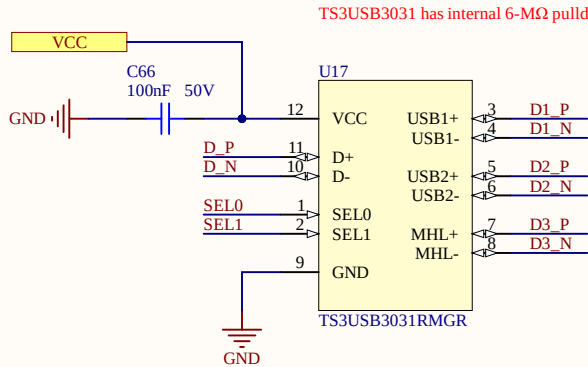
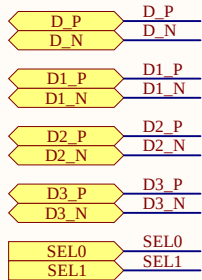
B

C

C

D

D



TS3USB3031 has internal 6-MΩ pulldown resistors on SEL0 and SEL1

Table 1. Function Table

SEL1	SEL0	SWITCH STATUS
Low	Low	D+/D- connected to USB1+/USB1-
Low	High	D+/D- connected to USB2+/USB2-
High	Low	D+/D- connected to MHL+/MHL-
High	High	USB and MHL switches in High-Z

Title TS3USB3031 2-ch 2p3t Multiplexer			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801
Size: A4	Number:*	Revision: 0.1	
Date: 11/26/2020	Time: 4:04:13 PM	Sheet* of *	
File: TS3USB3031_2-ch_2p3t_Multiplexer.SchDoc			

A

B

C

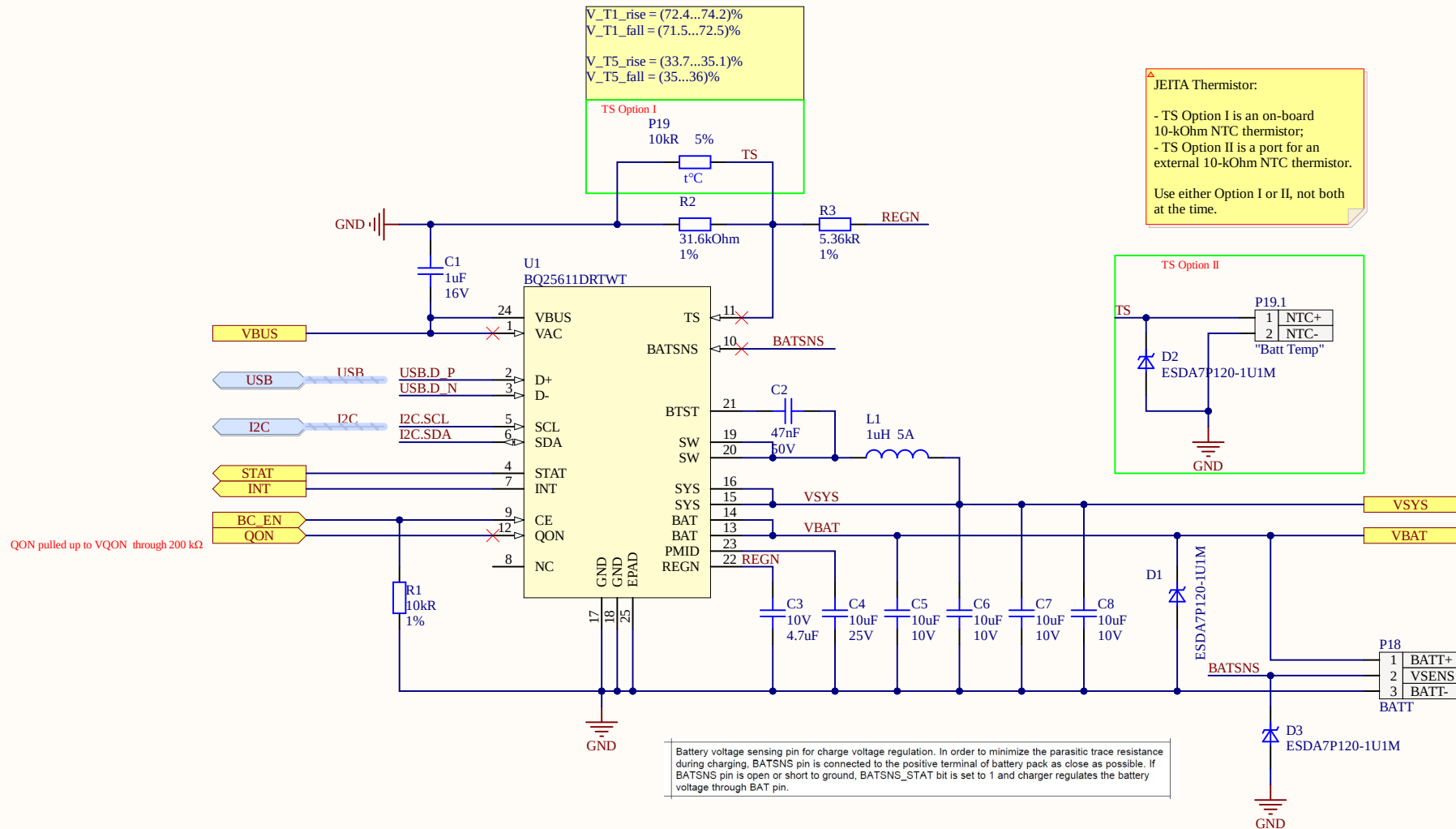
D

A

B

C

D

Title **bq25611d Battery Charger**Acme_CPU
30 N Gould St suite 4000
Sheridan WY 82801Size: **A4**Number: **3**Revision: **0.1**Date: **11/26/2020**Time: **4:04:14 PM**Sheet **3** of **50**File: **bq25611d_Battery_Charger.SchDoc**

A

A

B

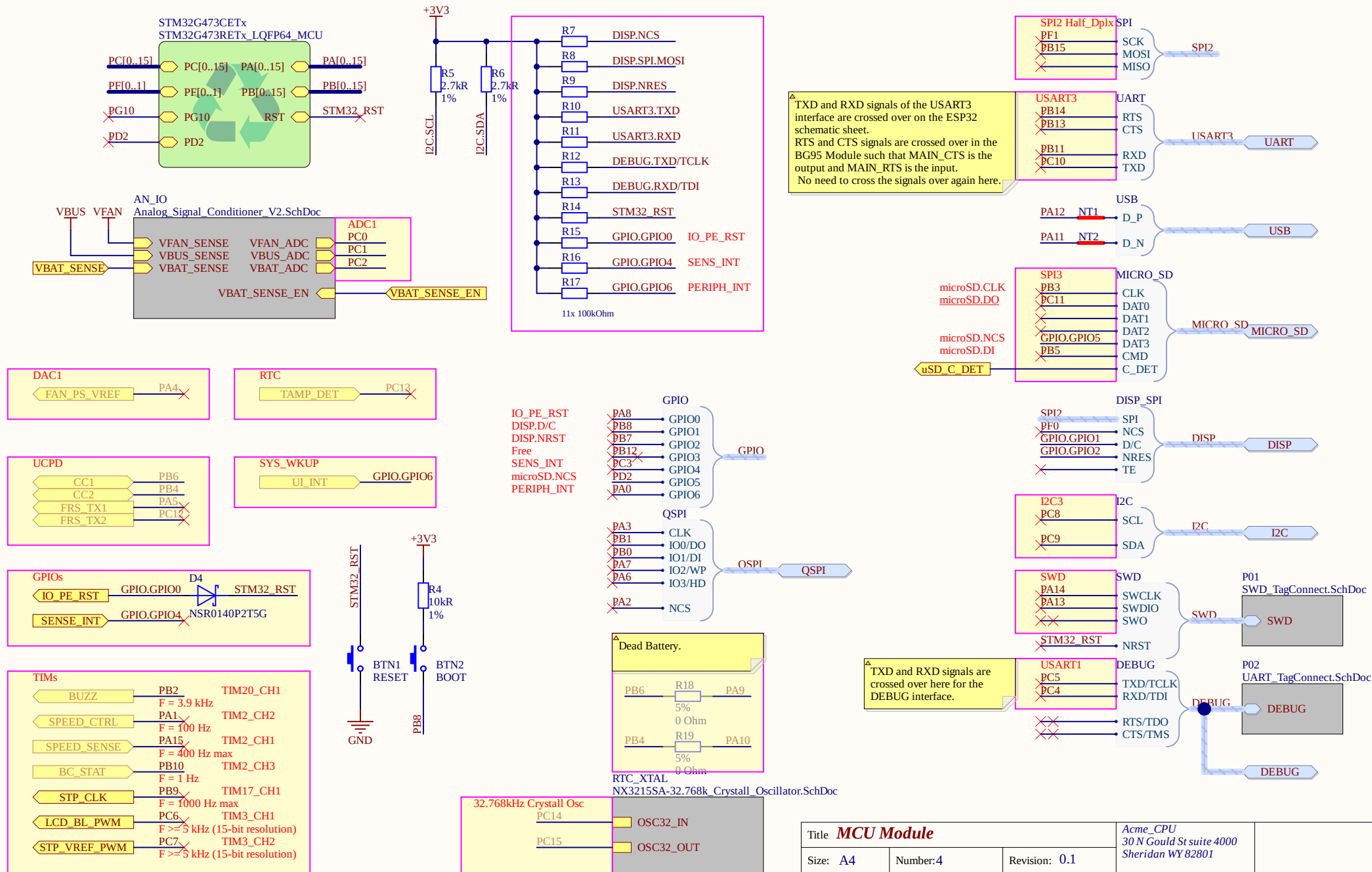
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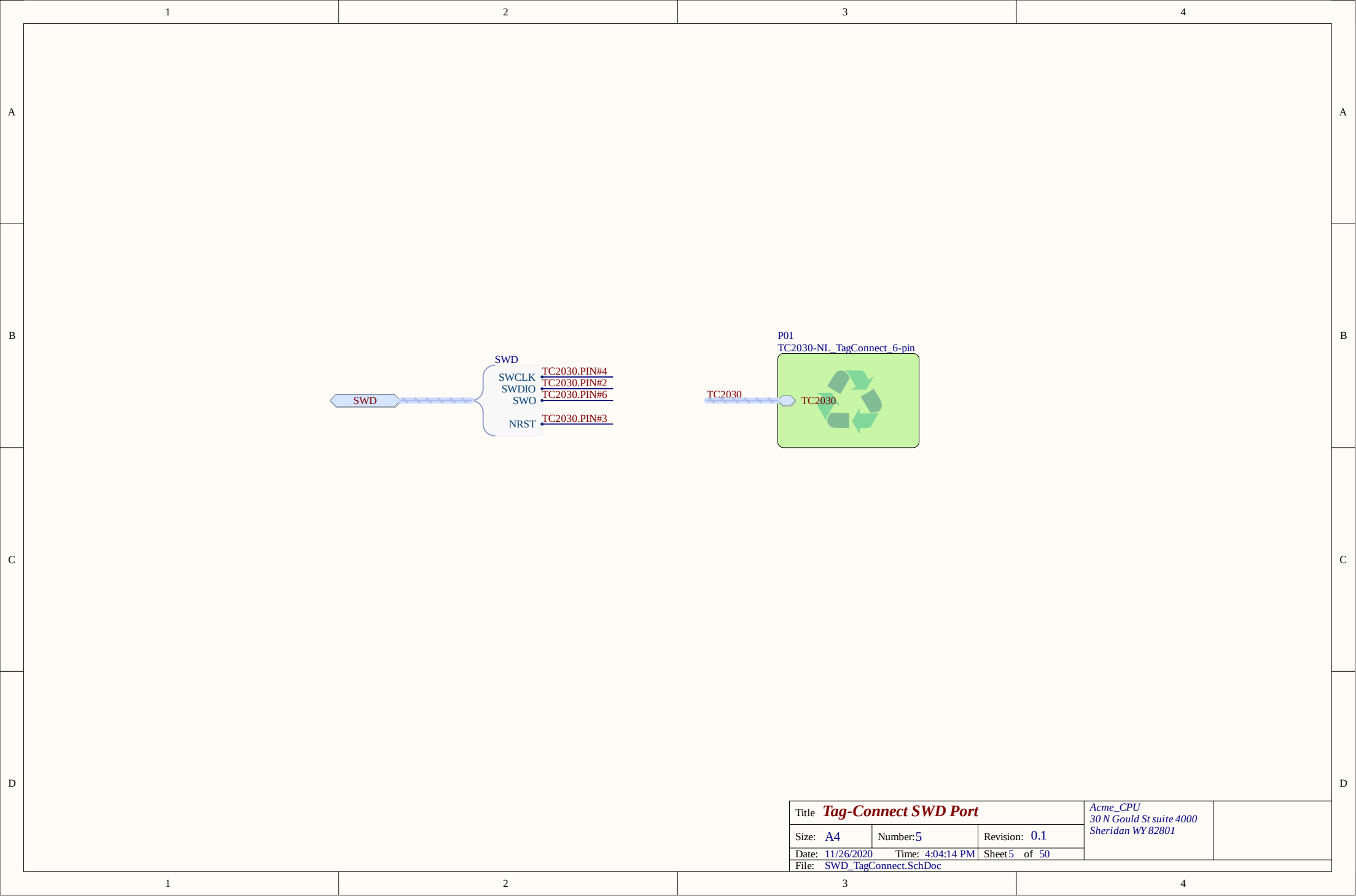
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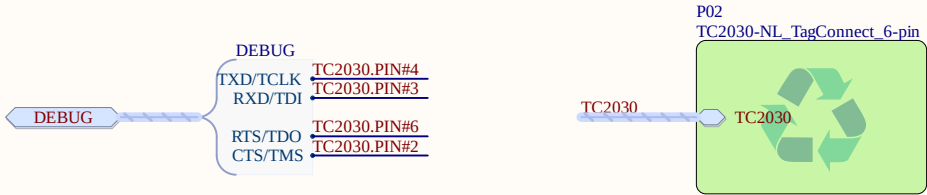
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D

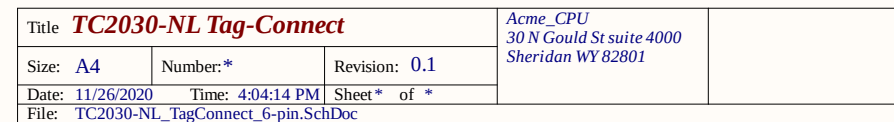
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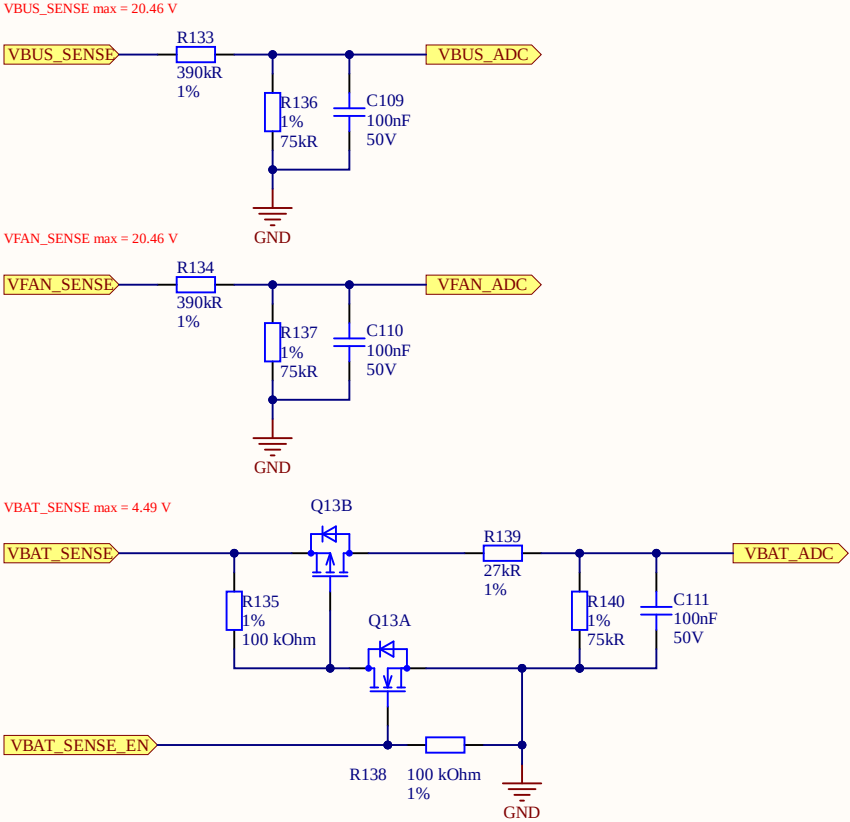
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30 N Gould St suite 4000
Sheridan WY 82801



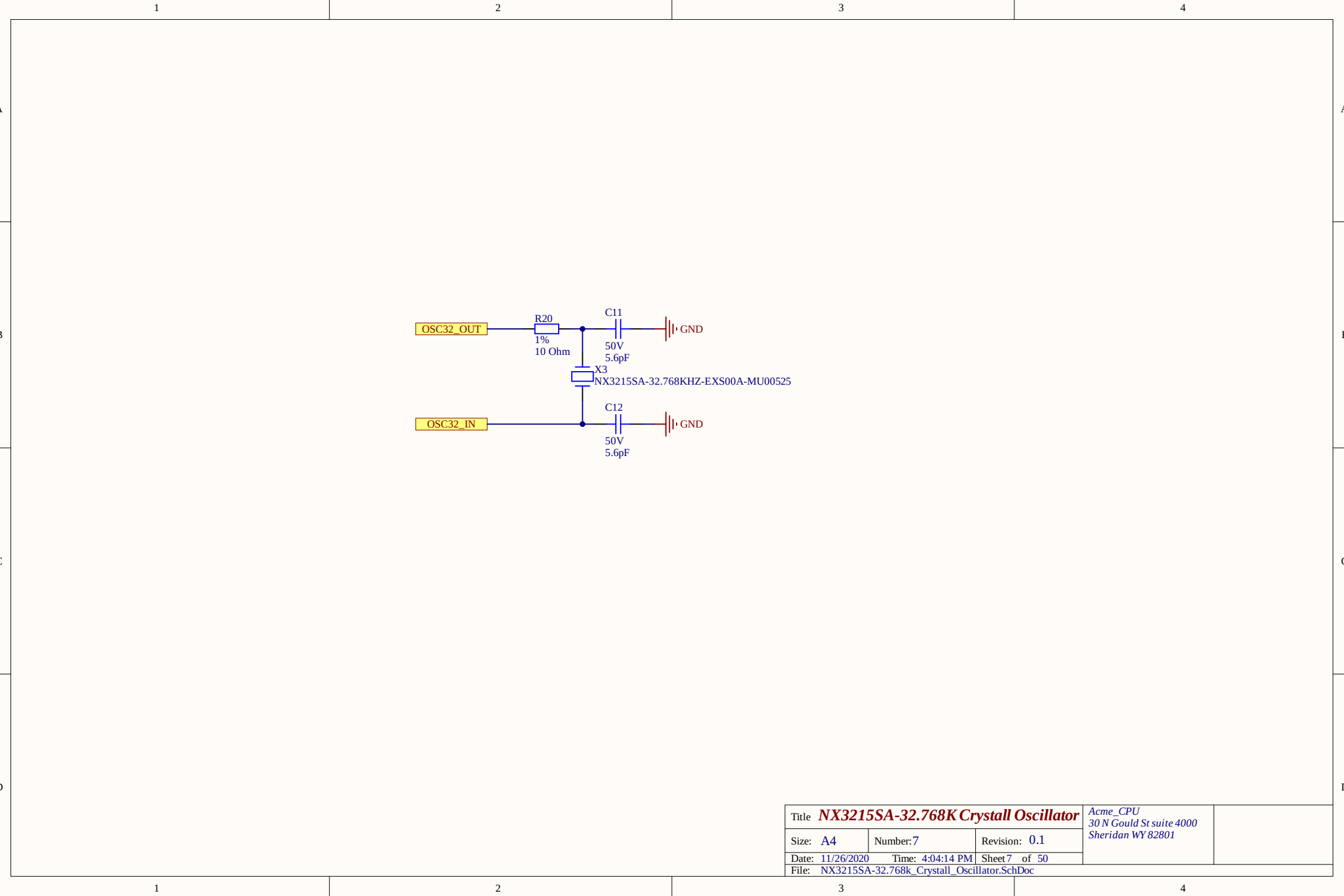


Title Tag-Connect DEBUG Port			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801
Size: A4	Number:6	Revision: 0.1	
Date: 11/26/2020	Time: 4:04:14 PM	Sheet 6 of 50	
File: UART_TagConnect.SchDoc			





Title Analog Signal Conditioner			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801		
Size: A4	Number: 8	Revision: 0.1			
Date: 11/26/2020	Time: 4:04:14 PM	Sheet 8 of 50			
File: Analog_Signal_Conditioner_V2.SchDoc					



A

B

C

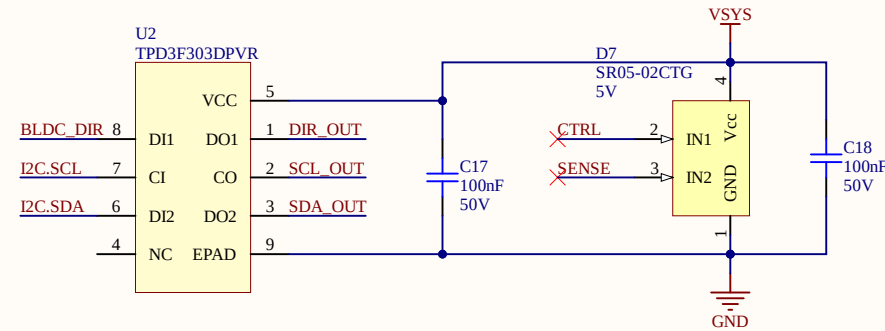
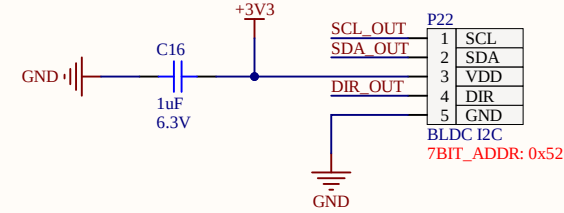
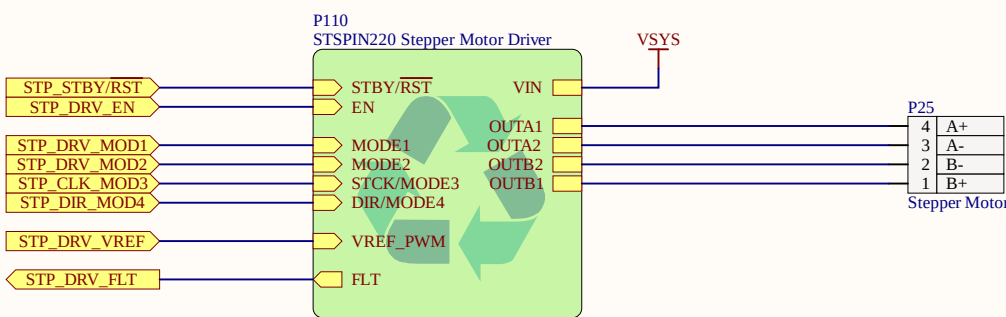
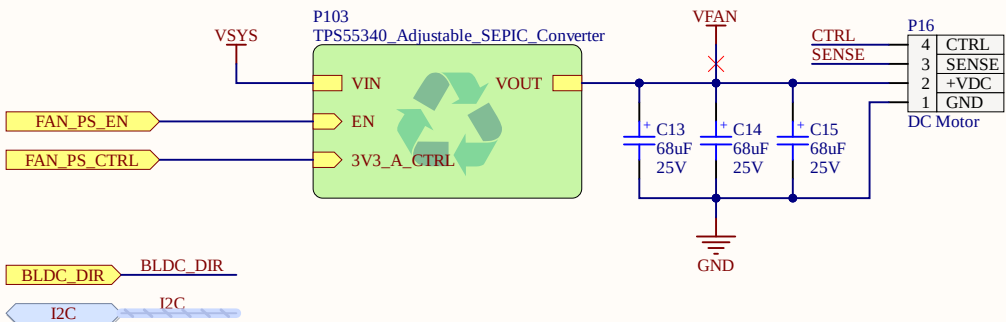
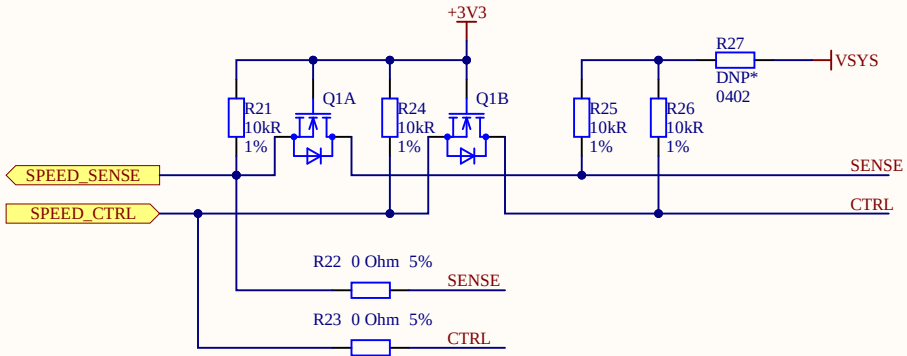
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A

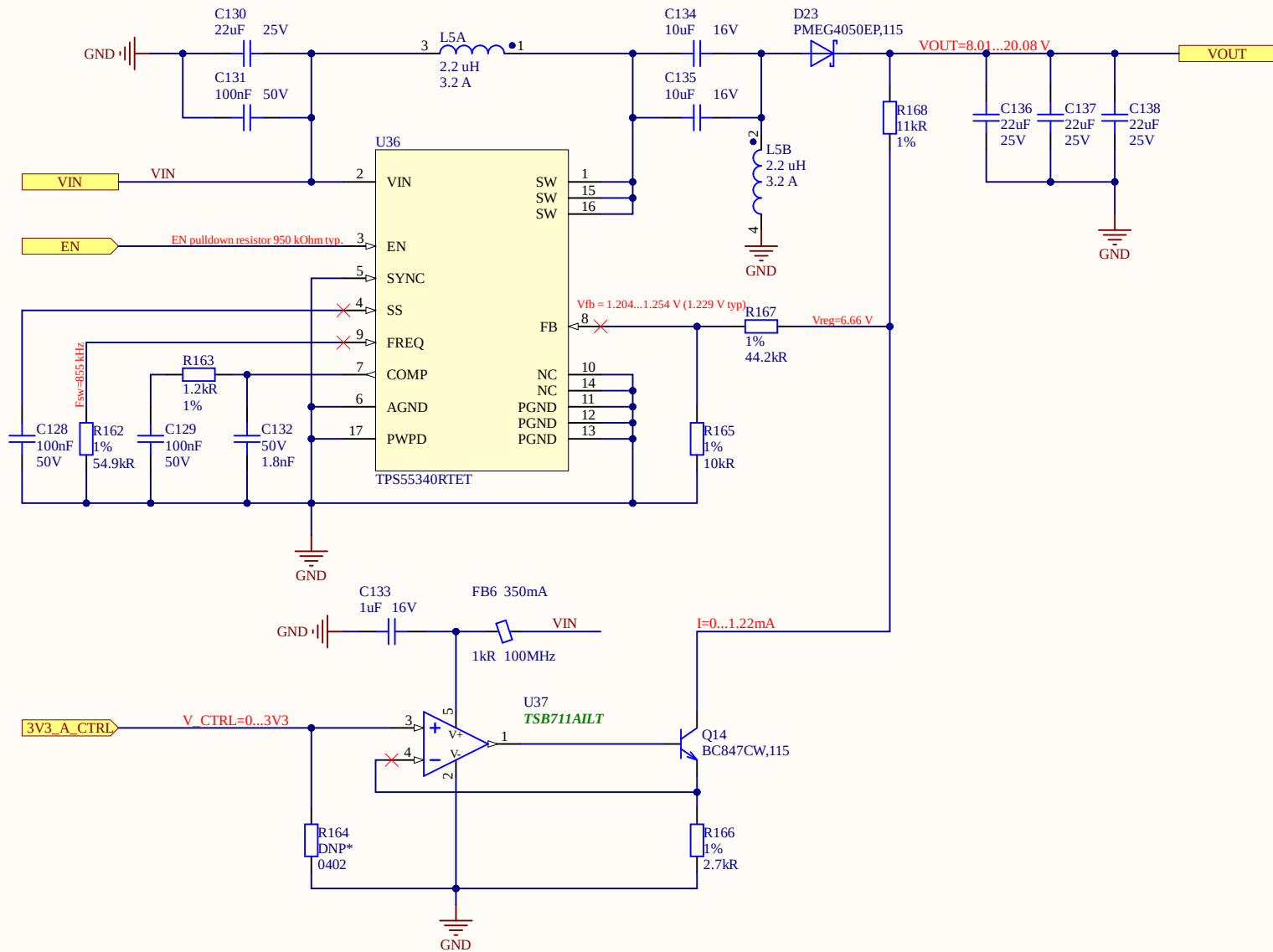
B

C

D



Title Motor Connection			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number: 9	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:14 PM	Sheet 9 of 50		
File: Motor Connection.SchDoc				



Title **+8V...20V Output SEPIC Converter**

Size: **A4**

Number: *

Revision: **0.1**

Date: **11/26/2020**

Time: **4:04:15 PM**

Sheet * of *

File: **TPS55340_Adjustable_SEPIC_Converter.SchDoc**

Acme_CPU
30 N Gould St suite 4000
Sheridan WY 82801

A

B

C

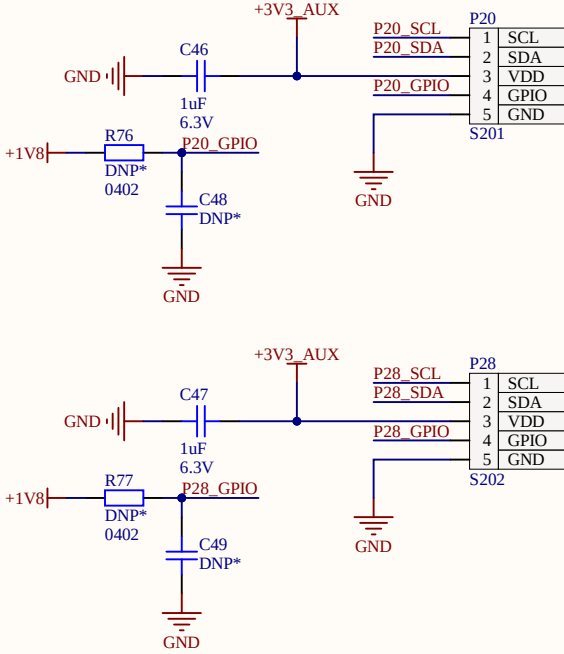
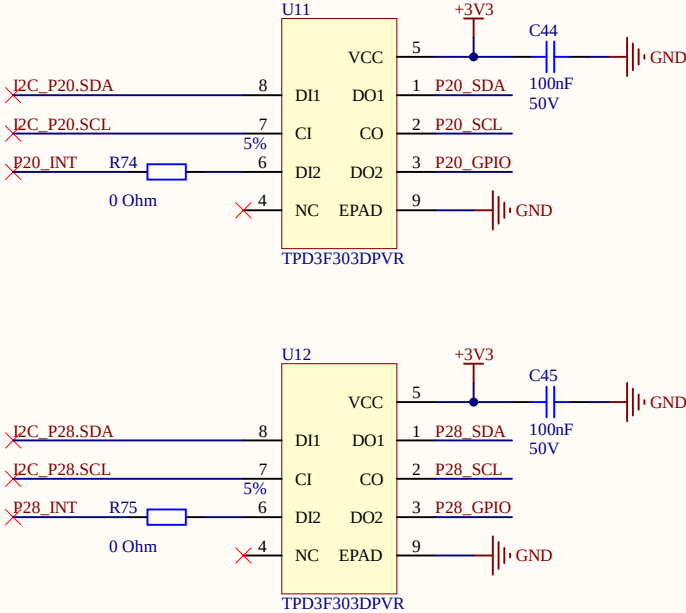
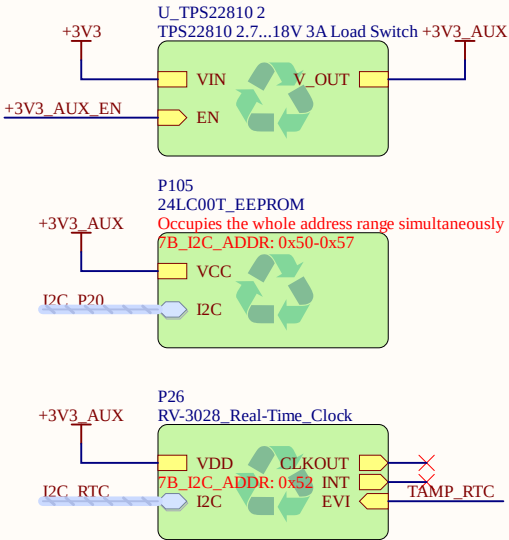
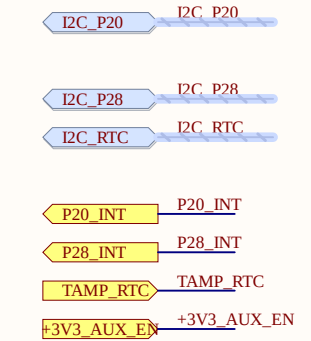
D

A

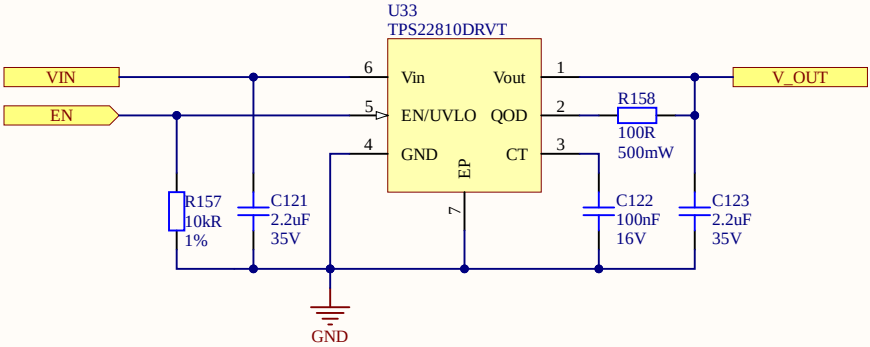
B

C

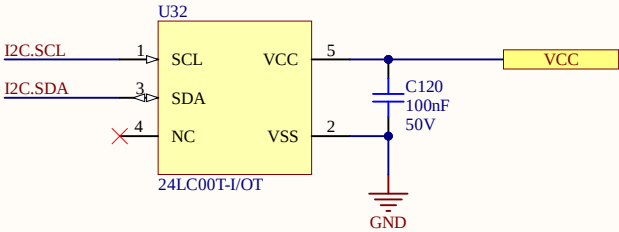
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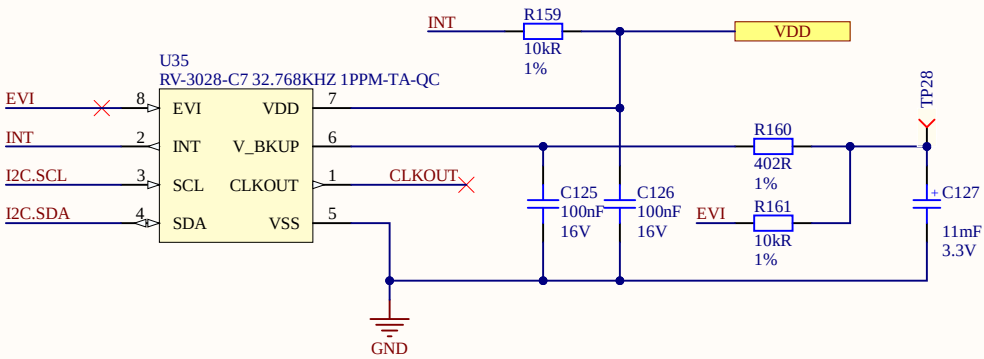
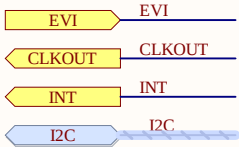
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Size: A4	Number: 10	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:15 PM	Sheet 10 of 50		
File: Sensor Connection.SchDoc				



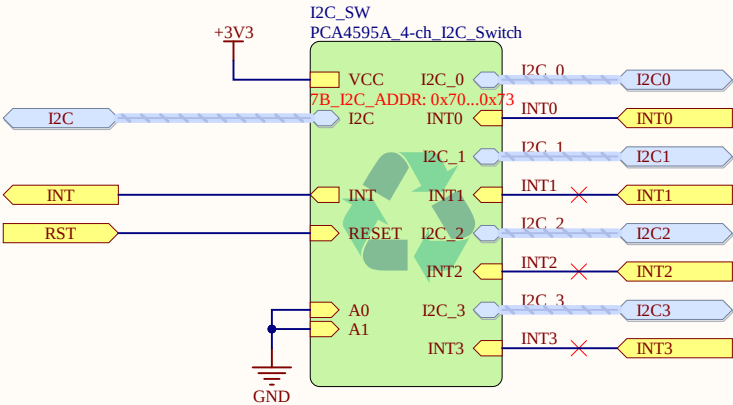
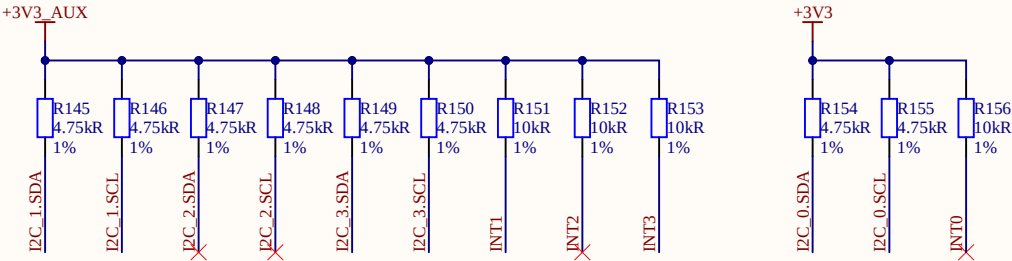
Title TPS22810 2.7...18V 3A Load Switch			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801
Size: A4	Number:*	Revision: 0.1	
Date: 11/26/2020	Time: 4:04:15 PM	Sheet* of *	
File: TPS22810 2.7...18V 3A Load Switch.SchDoc			



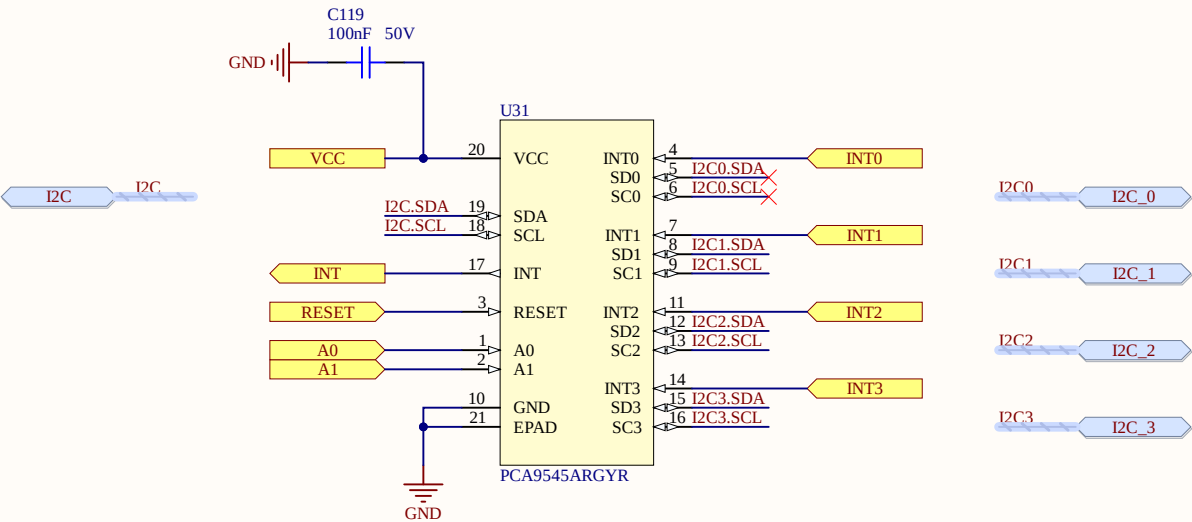
Title 24LC00T EEPROM			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number:*	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:15 PM	Sheet* of *		
File: 24LC00T_EEPROM.SchDoc				



Title RV-3028-C7 Real-Time Clock			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number:*	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:15 PM	Sheet* of *		
File: RV-3028_Real-Time_Clock.SchDoc				

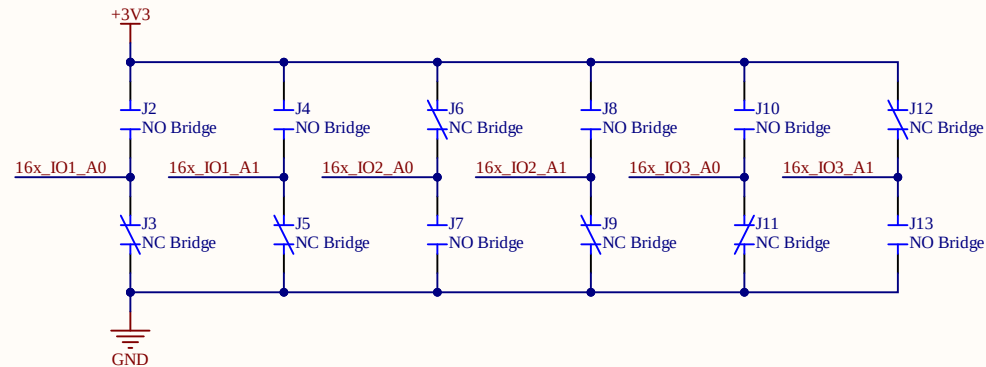
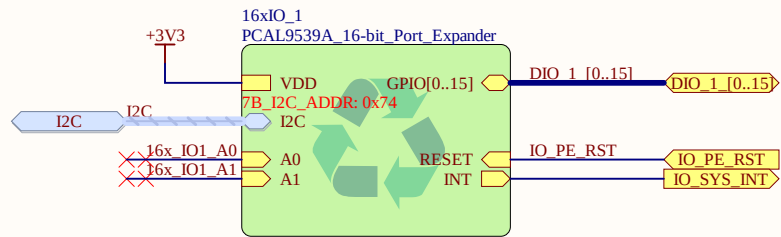


Title I2C Bus Multiplexer			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number: 11	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:15 PM	Sheet 11 of 50		
File: I2C_Bus_Multiplexer.SchDoc				



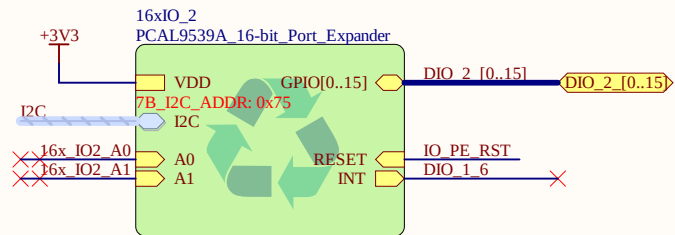
Title PCA9545A 4-ch I2C Switch			<i>Acme_CPU</i> <i>30 N Gould St suite 4000</i> <i>Sheridan WY 82801</i>		
Size: A4	Number: *	Revision: 0.1			
Date: 11/26/2020	Time: 4:04:16 PM	Sheet * of *			
File: PCA4595A_4-ch_I2C_Switch.SchDoc					

A



A

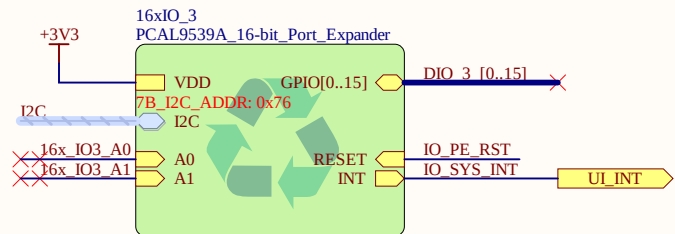
B



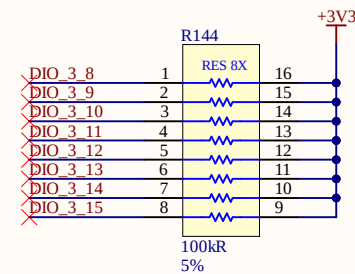
DIO_2_13
DIO_2_14

B

C



DIO_3 [0..7] DIO_3 [8..15]
DIO_3 [0..7] UI_IO [8..15]

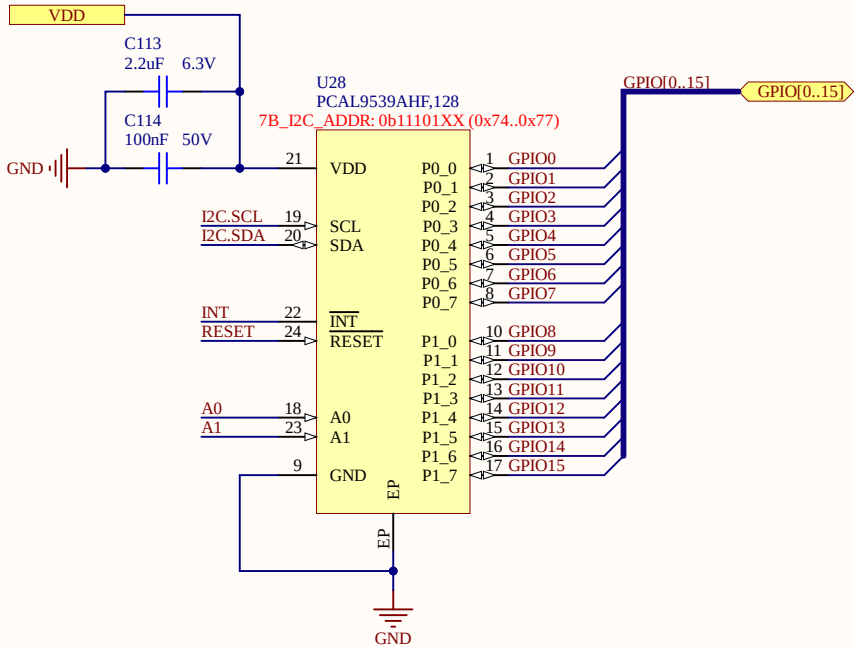
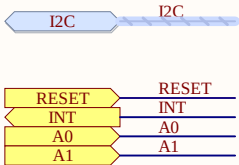


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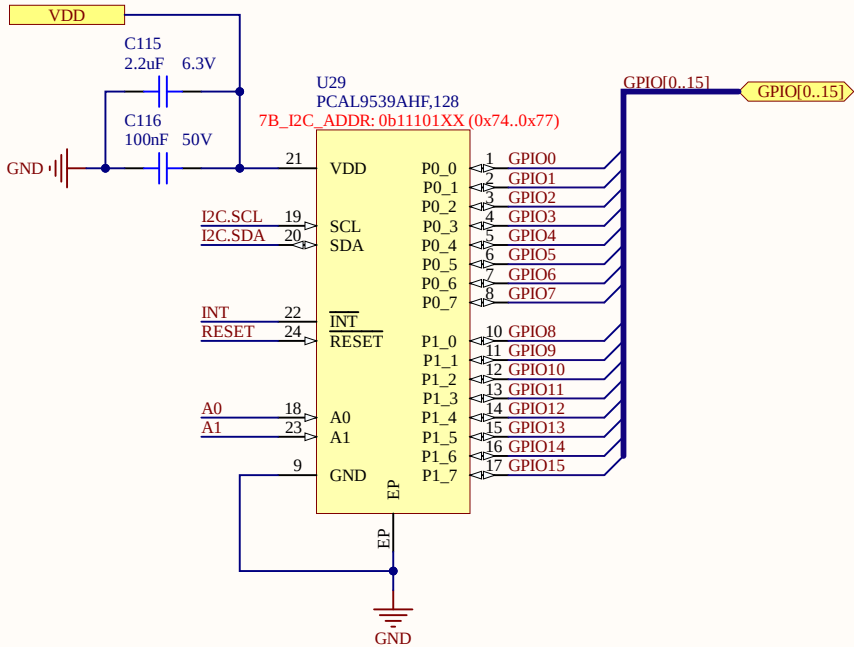
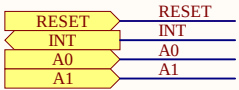
D

Title Logic IO System			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number: 12	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:16 PM	Sheet 12 of 50		
File: Logic_IO_Unit_V2.SchDoc				

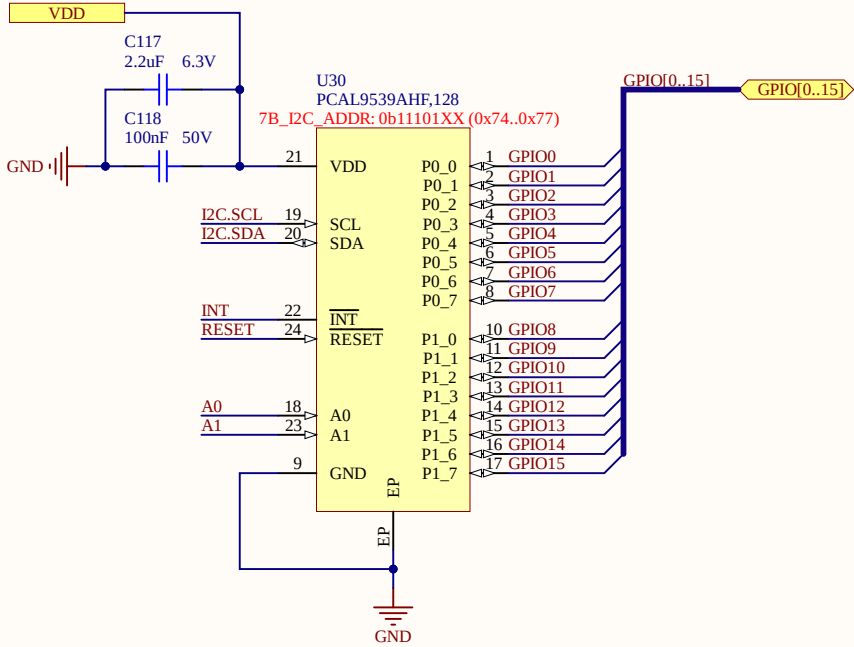
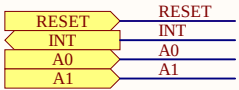
D



Title PCAL9539A 16-bit Port Expander			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801
Size: A4	Number:*	Revision: 0.1	
Date: 11/26/2020	Time: 4:04:16 PM	Sheet* of *	
File: PCAL9539A_16-bit_Port_Expander.SchDoc			



Title PCAL9539A 16-bit Port Expander			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801
Size: A4	Number:*	Revision: 0.1	
Date: 11/26/2020	Time: 4:04:16 PM	Sheet* of *	
File: PCAL9539A_16-bit_Port_Expander.SchDoc			



Title PCAL9539A 16-bit Port Expander			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number:*	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:16 PM	Sheet* of *		
File: PCAL9539A_16-bit_Port_Expander.SchDoc				

A

A

B

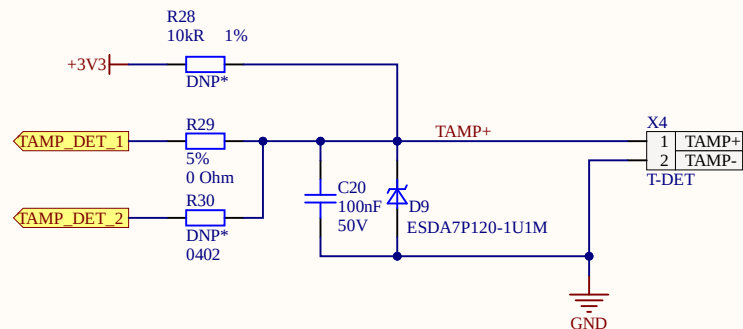
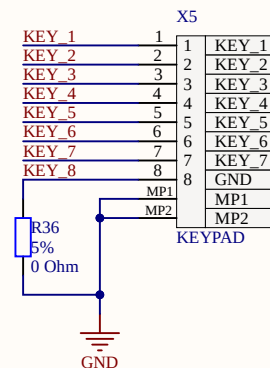
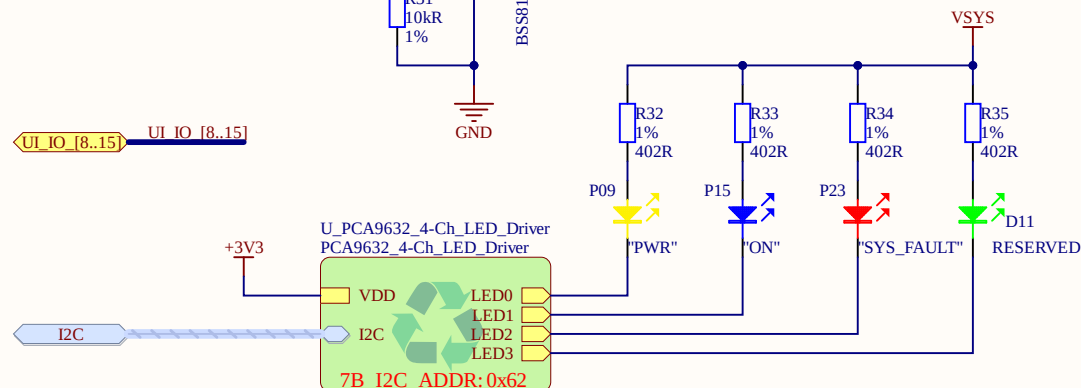
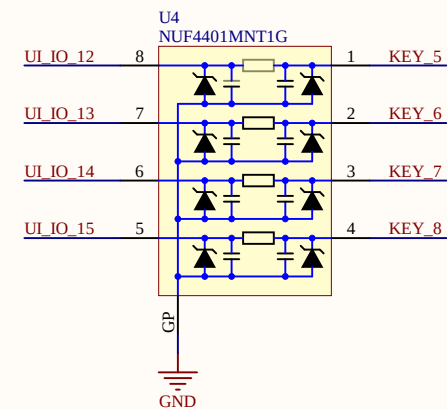
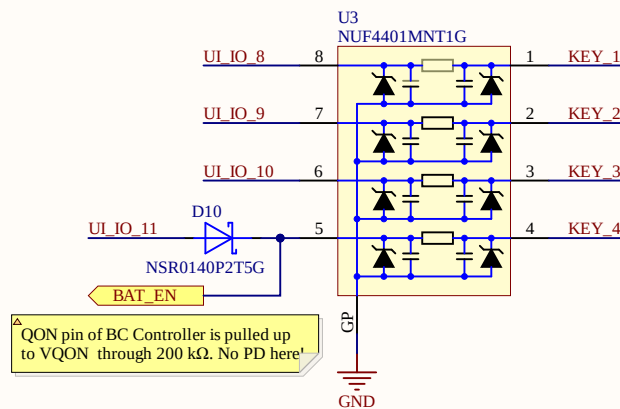
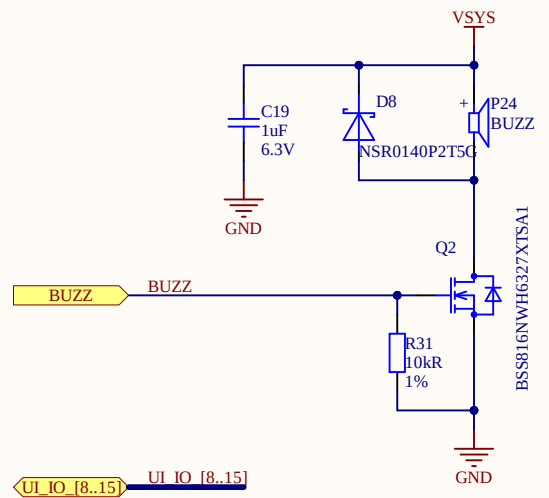
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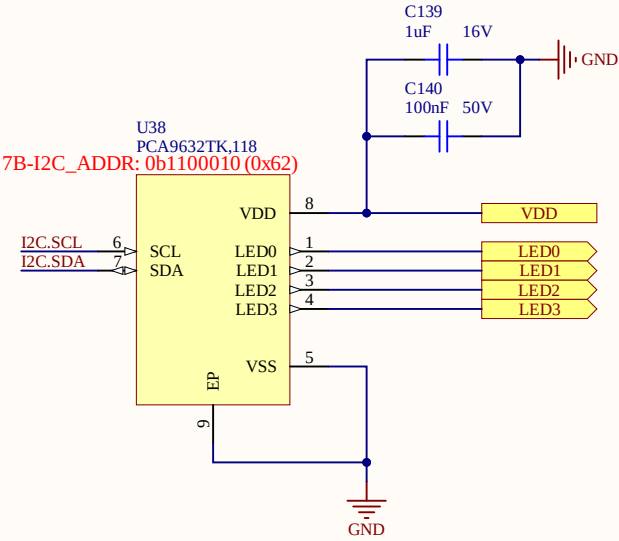
C

C

D

D

Title **User Interface**Size: **A4**Number: **13**Revision: **0.1**Date: **11/26/2020**Time: **4:04:16 PM**Sheet **13** of **50**File: **User_Interface.SchDoc****Acme_CPU**
30 N Gould St suite 4000
Sheridan WY 82801



Each LED output can be off, on (no PWM control), set at its Individual PWM controller value or at both Individual and Group PWM controller values. The LED output driver is programmed to be either open-drain with a 25 mA current sink capability at 5 V or totem pole with a 25 mA sink, 10 mA source capability at 5 V. The PCA9632 operates with a supply voltage range of 2.3 V to 5.5 V and the outputs are 5.5 V tolerant.

Title PCA9632 4-ch LED Driver			<i>Acme_CPU</i> <i>30 N Gould St suite 4000</i> <i>Sheridan WY 82801</i>	
Size: A4	Number:*	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:16 PM	Sheet * of *		
File: PCA9632_4-Ch_LED_Driver.SchDoc				

A

B

A

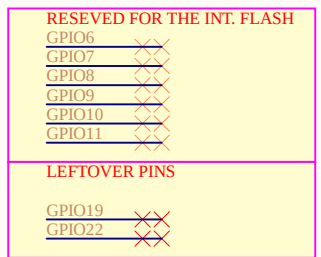
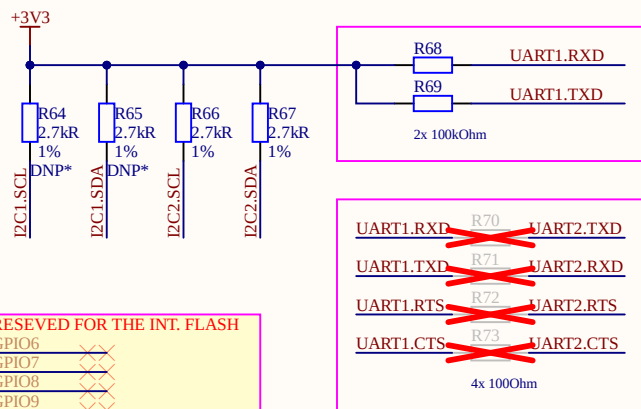
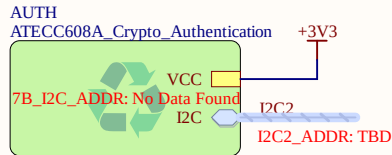
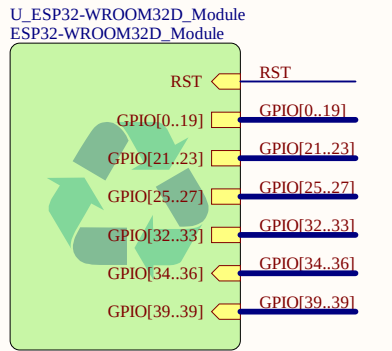
B

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C

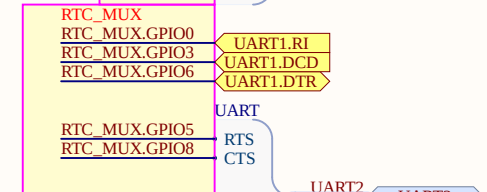
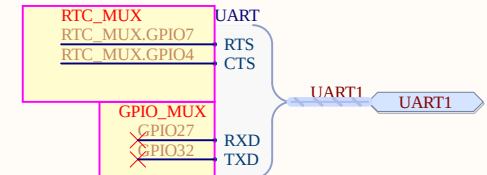
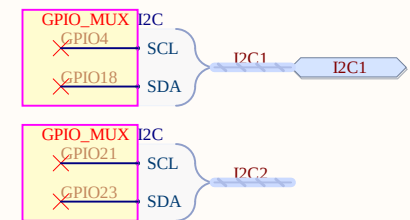
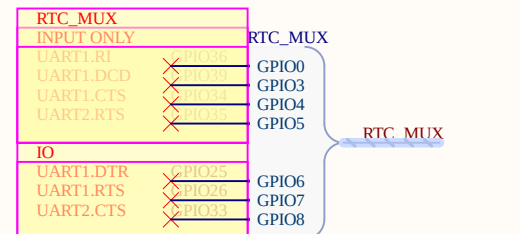
D

D

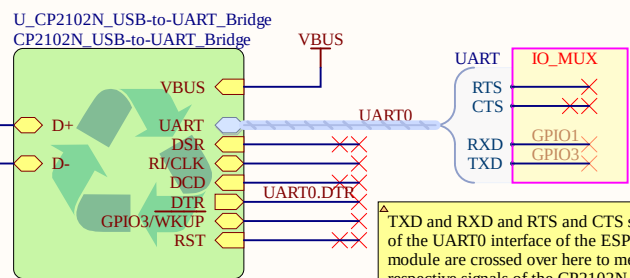
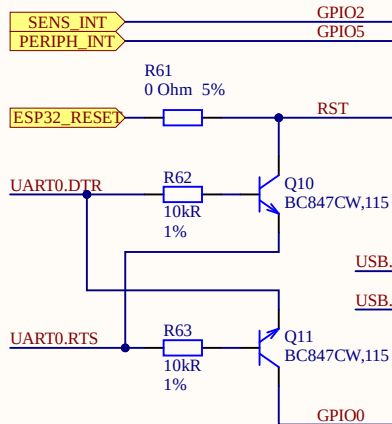
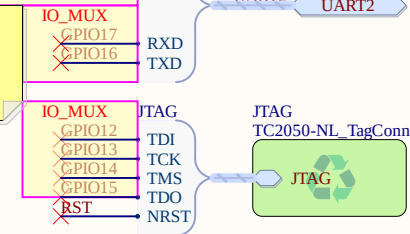


A TXD and RXD signals of the UART1 and UART2 interfaces of the ESP32 module are crossed over here so that if ESP32 Module is not installed and the STM32 MCU is connected to the BG95 Module through the resistor array, the TXD and RXD signals of the STM32 would meet respective RXD and TXD signals of the BG95 Broadband Modem.

RTS and CTS signals are crossed over in the BG95 Module so no need to cross them over again here.

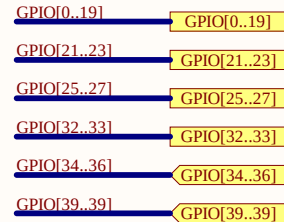
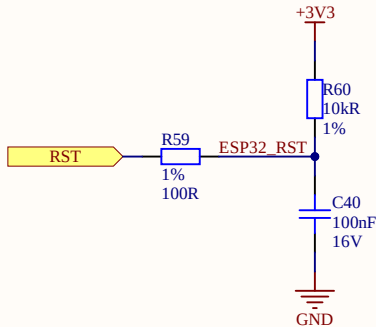


A TXD and RXD signals of the UART2 interface of the ESP32 module are crossed over here to meet respective signals of the BG95 Broadband Modem.

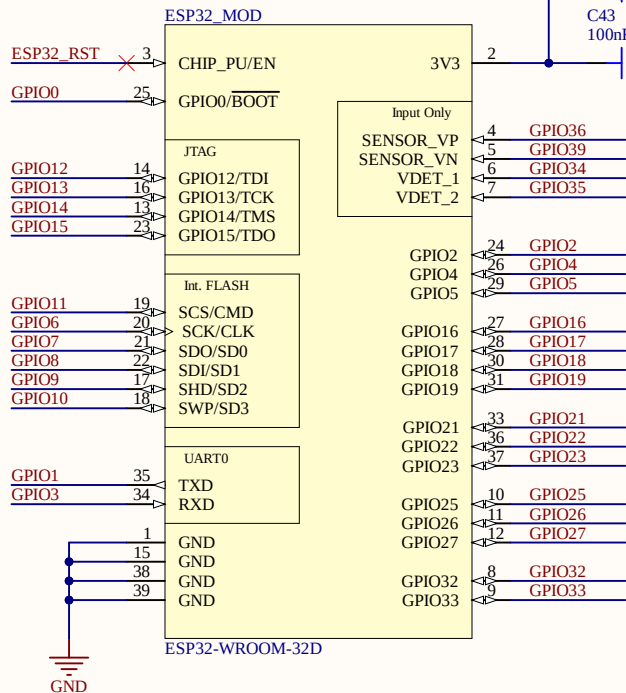


A TXD and RXD and RTS and CTS signals of the UART0 interface of the ESP32 module are crossed over here to meet respective signals of the CP2102N USB-UART bridge.

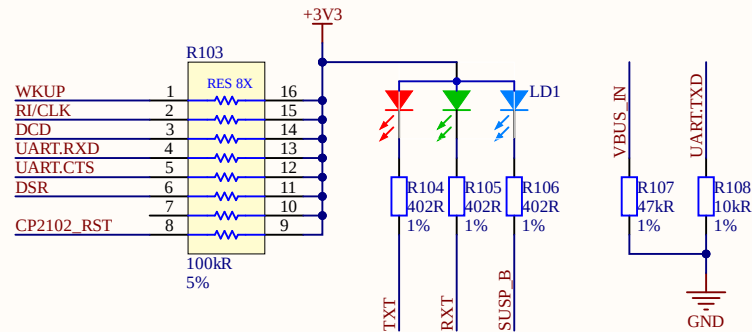
Title ESP32-WROOM32D Module			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number: 14	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:16 PM	Sheet 14 of 50		
File: ESP32_Module.SchDoc				



Pins SCK/CLK, SDO/SD0, SDI/SD1, SHD/SD2, SWP/SD3 and SCS/CMD, namely, GPIO6 to GPIO11 are connected to the integrated SPI flash integrated on the module and are not recommended for other uses.

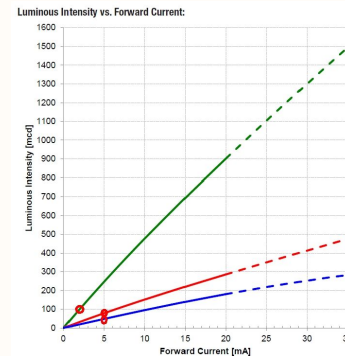


Title ESP32-WROOM32D Module			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801		
Size: A4	Number: *	Revision: 0.1			
Date: 11/26/2020	Time: 4:04:17 PM	Sheet * of *			
File: ESP32-WROOM32D_Module.SchDoc					



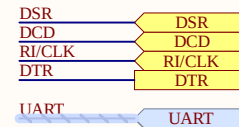
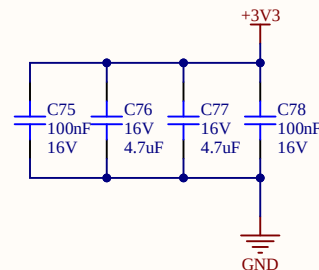
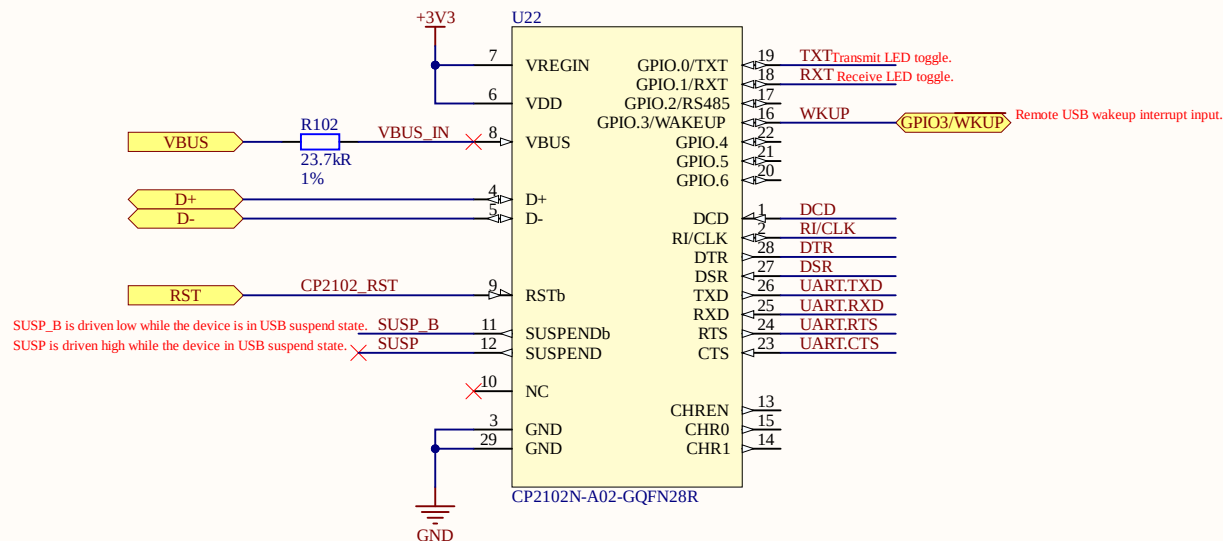
4.2.2 Sending Break Signaling

The CP2102N supports break signaling with an external 10k Ohm resistor between TXD and ground. This resistor is sufficient for break signaling across all baud rates.

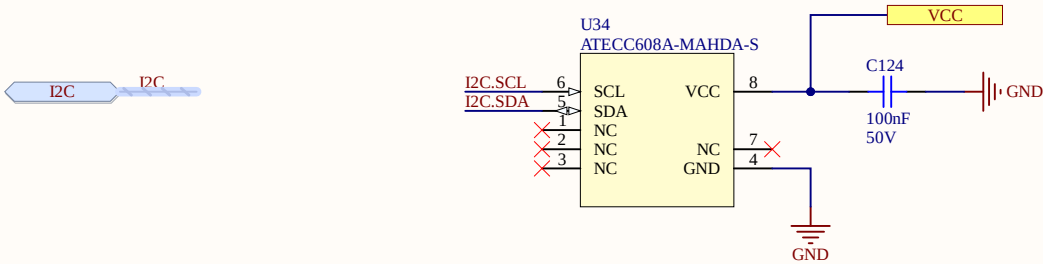


LED's current
Green - 2 mA or less;
Red & Blue - approx. 3-5 mA.

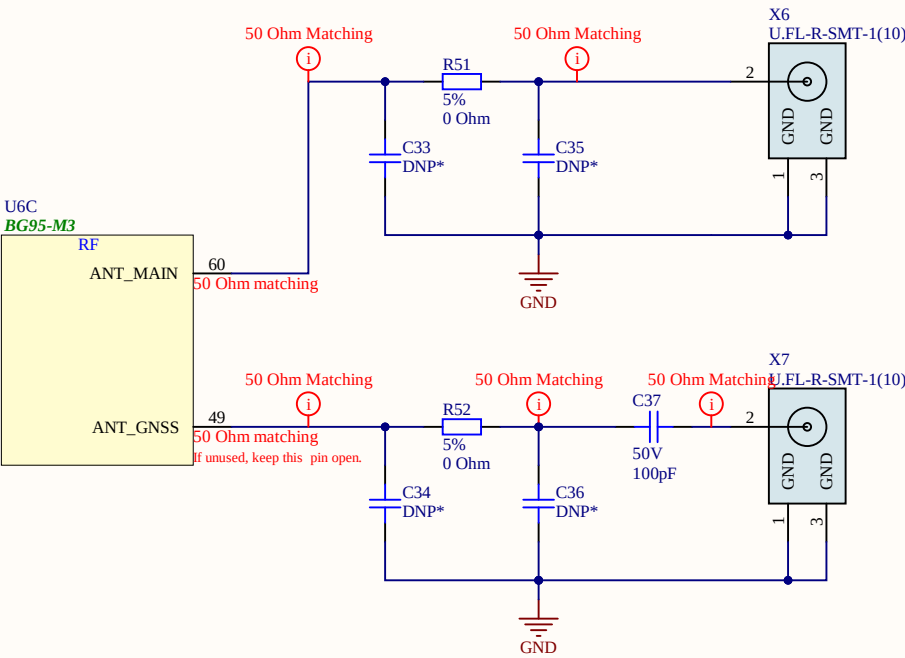
Forward Voltage @ desired current:
Red - 1.8 V;
Green & Blue - approx. 2.8 V.



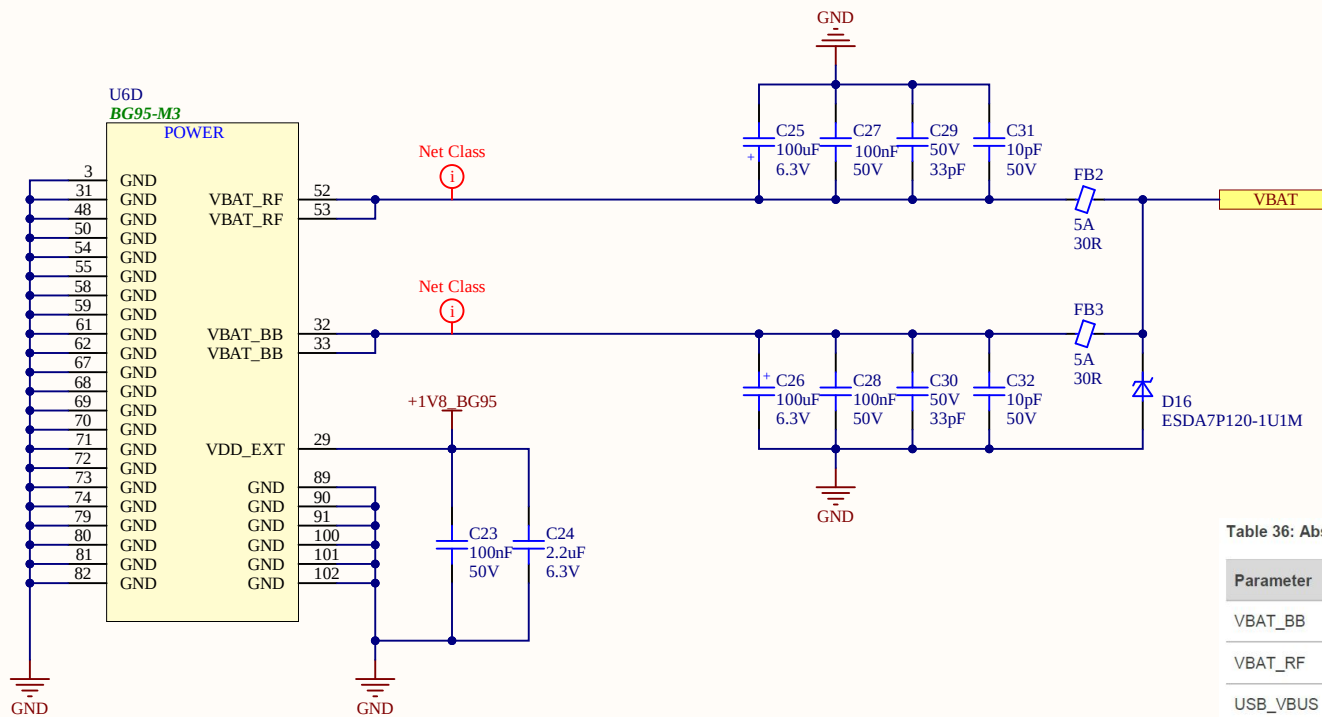
Title CP2102 USB-to-UART Bridge			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number: *	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:17 PM	Sheet * of *		
File: CP2102N_USB-to-UART_Bridge.SchDoc				



Title ATECC608A Crypto Co-Processor			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801
Size: A4	Number:*	Revision: 0.1	
Date: 11/26/2020	Time: 4:04:17 PM	Sheet* of *	
File: ATECC608A_Crypto_Authentication.SchDoc			



Title BG95 RF Interface			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number: 16	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:17 PM	Sheet 16 of 50		
File: BG95_RF.SchDoc				



A

A

B

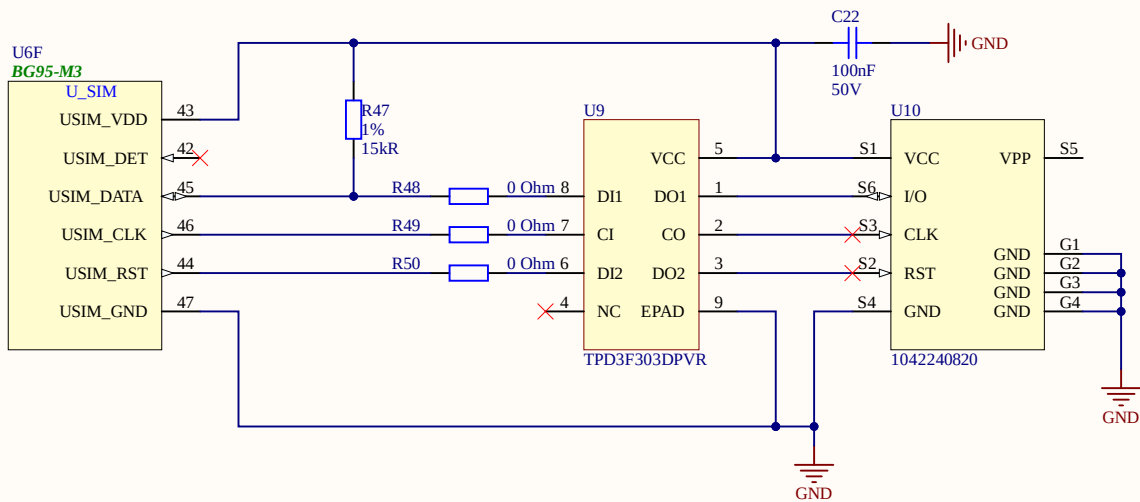
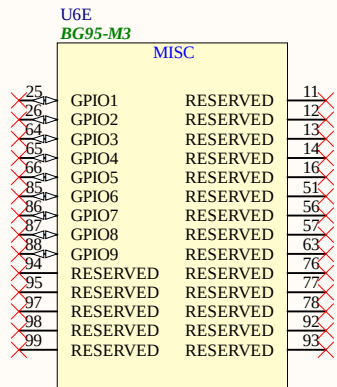
B

C

C

D

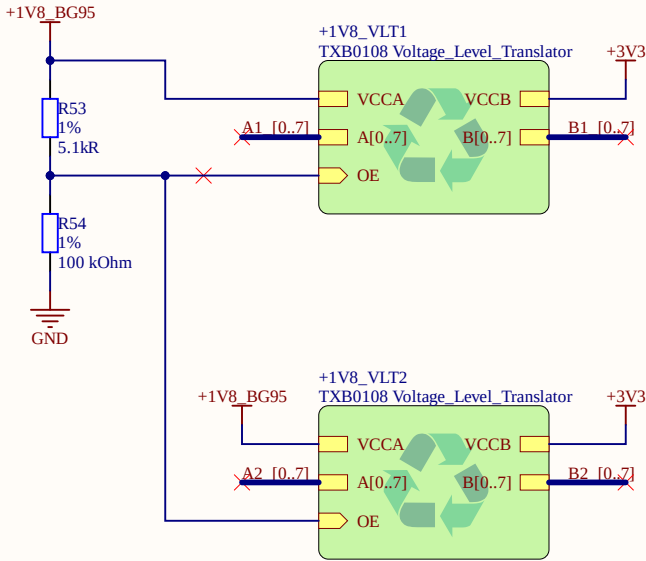
D



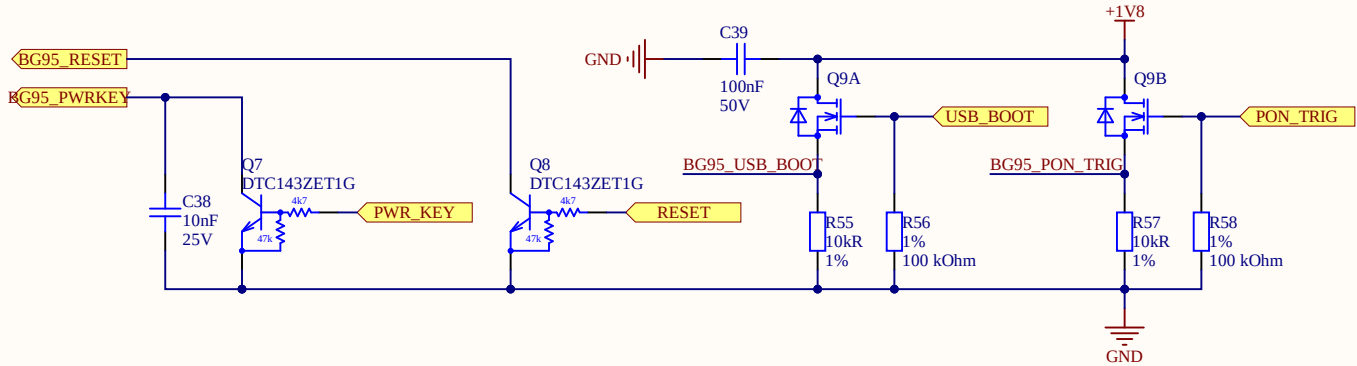
Title BG95_Misc			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number: 18	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:17 PM	Sheet 18 of 50		
File: BG95_Misc.SchDoc				

A0	A1_0
A1	A1_1
A2	A1_2
A3	A1_3
A4	A1_4
A5	A1_5
A6	A1_6
A7	A1_7
A8	A2_0
A9	A2_1
A10	A2_2
A11	A2_3
A12	A2_4
A13	A2_5
A14	A2_6
A15	A2_7

B1_0	B0
B1_1	B1
B1_2	B2
B1_3	B3
B1_4	B4
B1_5	B5
B1_6	B6
B1_7	B7
B2_0	B8
B2_1	B9
B2_2	B10
B2_3	B11
B2_4	B12
B2_5	B13
B2_6	B14
B2_7	B15



BG95_PON_TRIG	BG95_PON_TRIG
BG95_USB_BOOT	BG95_USB_BOOT



A

B

C

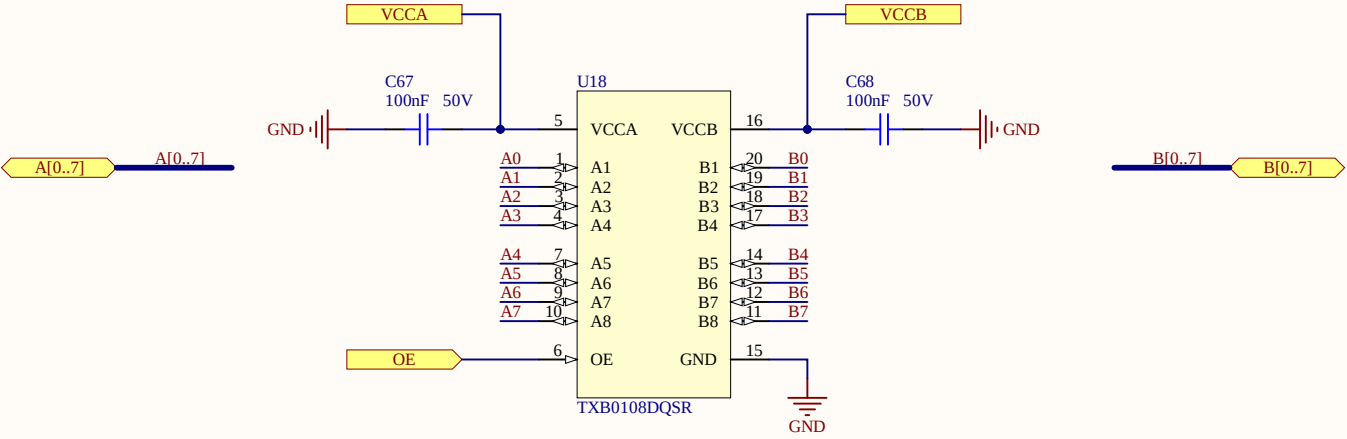
D

A

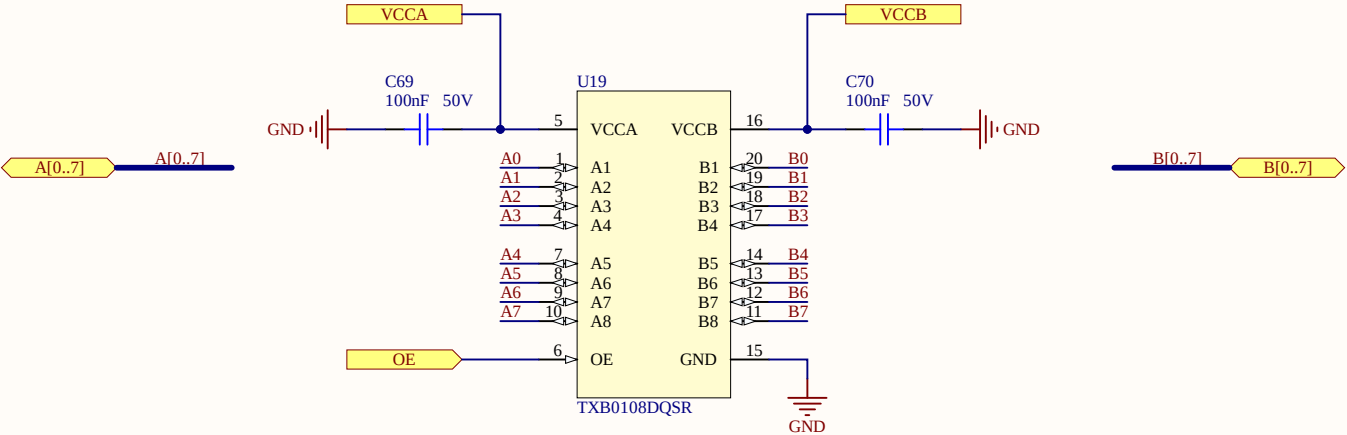
B

C

D



Title TXB0108 Voltage Level Transator			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801
Size: A4	Number:*	Revision: 0.1	
Date: 11/26/2020	Time: 4:04:18 PM	Sheet* of *	
File: TXB0108 Voltage_Level_Translator.SchDoc			



Title TXB0108 Voltage Level Transator			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801
Size: A4	Number:*	Revision: 0.1	
Date: 11/26/2020	Time: 4:04:18 PM	Sheet* of *	
File: TXB0108 Voltage_Level_Translator.SchDoc			

A

A

B

B

C

C

D

D

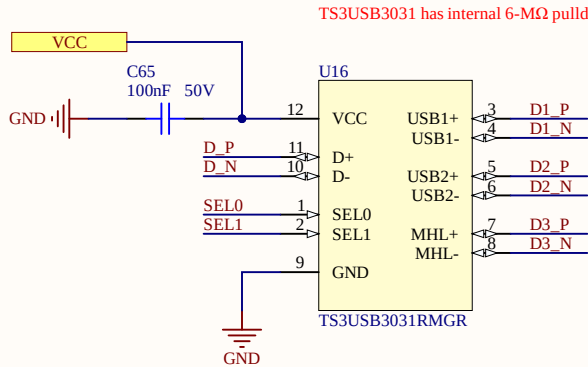
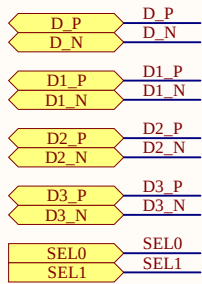
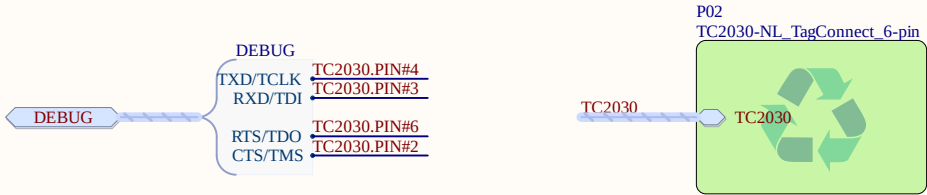


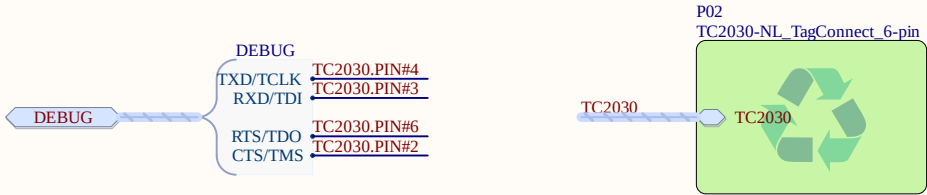
Table 1. Function Table

SEL1	SEL0	SWITCH STATUS
Low	Low	D+/D- connected to USB1+/USB1-
Low	High	D+/D- connected to USB2+/USB2-
High	Low	D+/D- connected to MHL+/MHL-
High	High	USB and MHL switches in High-Z

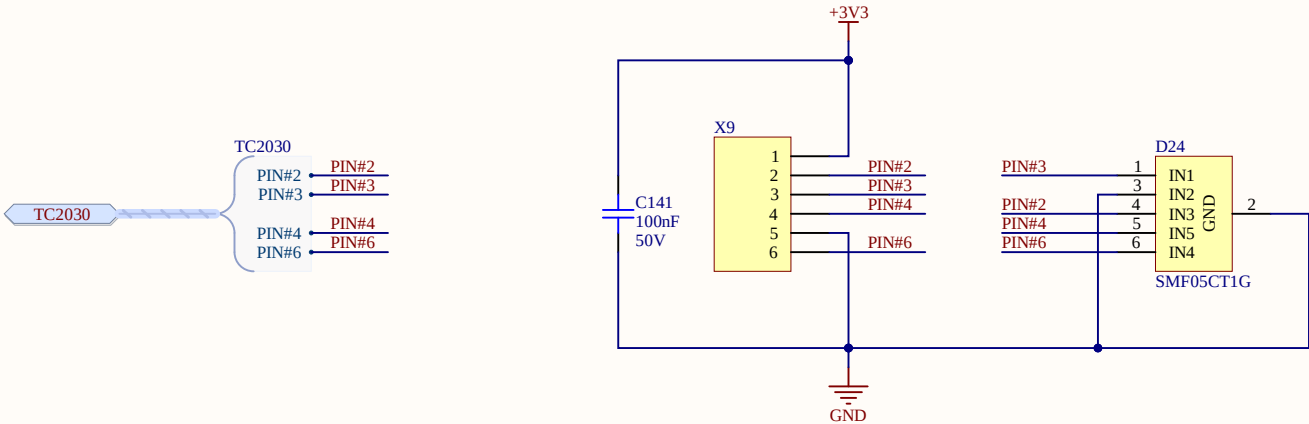
Title TS3USB3031 2-ch 2p3t Multiplexer			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801
Size: A4	Number:*	Revision: 0.1	
Date: 11/26/2020	Time: 4:04:18 PM	Sheet* of *	
File: TS3USB3031_2-ch_2p3t_Multiplexer.SchDoc			



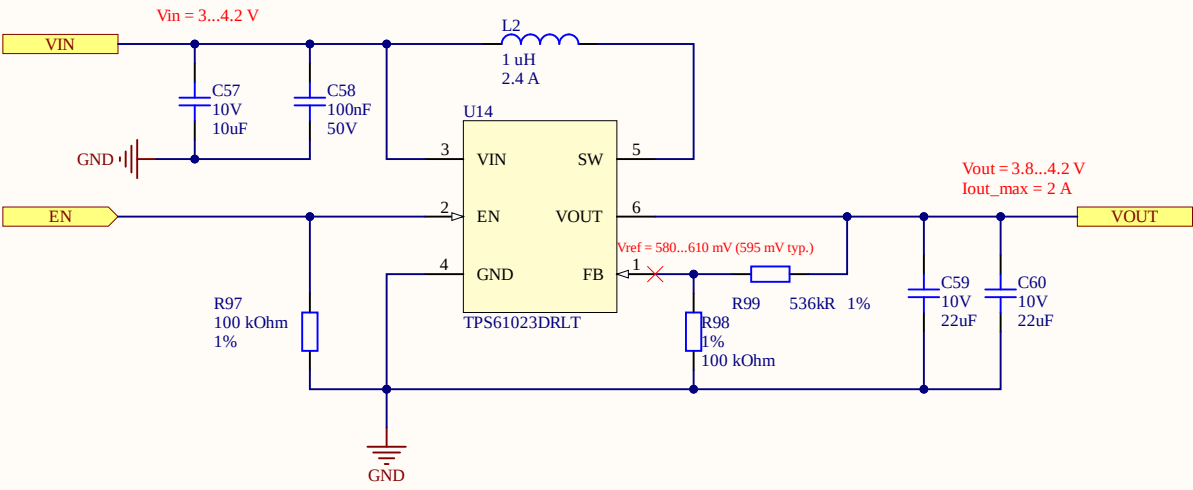
Title Tag-Connect DEBUG Port			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801
Size: A4	Number: 6	Revision: 0.1	
Date: 11/26/2020	Time: 4:04:18 PM	Sheet 6 of 50	
File: UART_TagConnect.SchDoc			



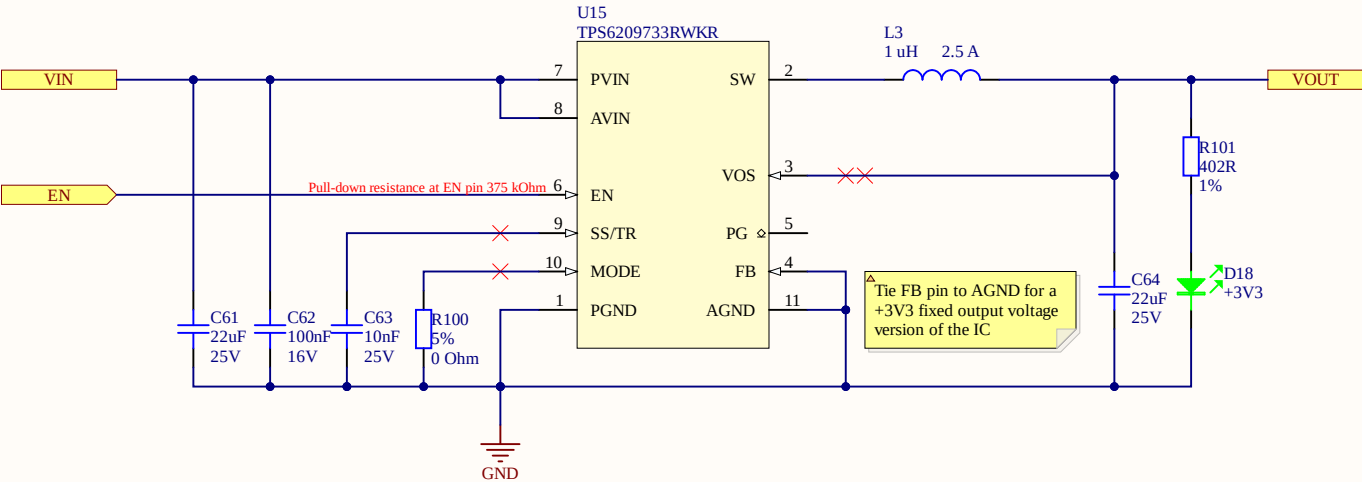
Title Tag-Connect DEBUG Port			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801
Size: A4	Number: 6	Revision: 0.1	
Date: 11/26/2020	Time: 4:04:18 PM	Sheet 6 of 50	
File: UART_TagConnect.SchDoc			



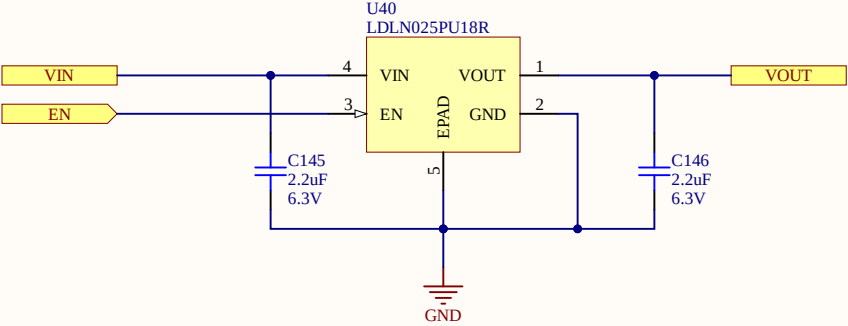
Title TC2030-NL Tag-Connect			<i>Acme_CPU</i> 30 N Gould St suite 4000 Sheridan WY 82801		
Size: A4	Number:*	Revision: 0.1			
Date: 11/26/2020	Time: 4:04:18 PM	Sheet* of *			
File: TC2030-NL_TagConnect_6-pin.SchDoc					



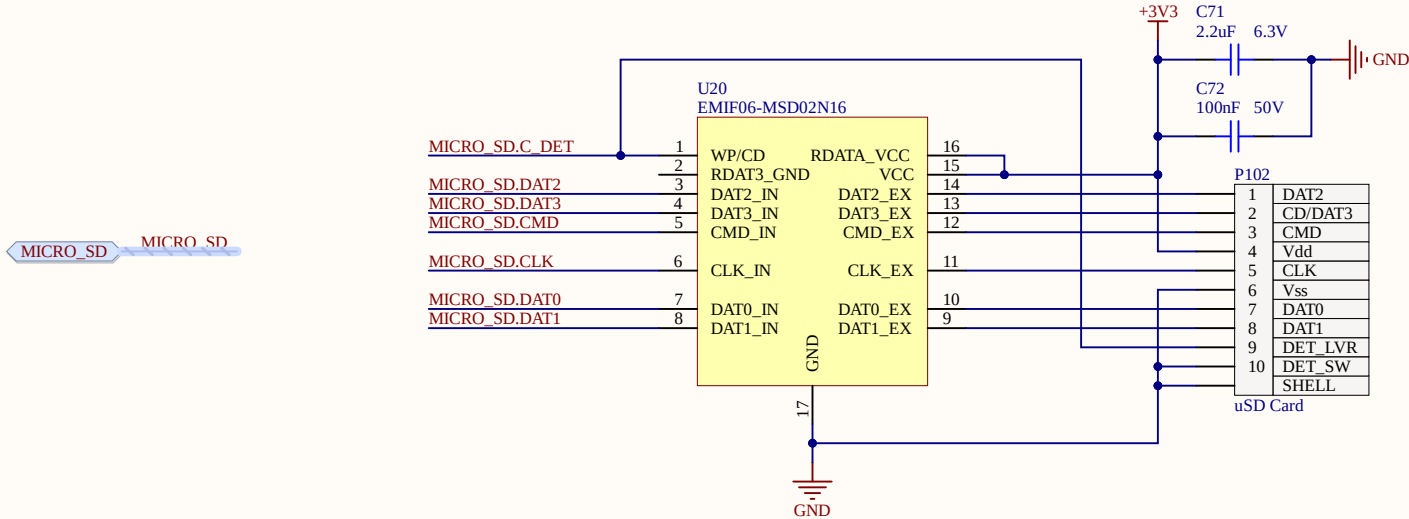
Title TPS61023 Boost Converter			<i>Acme_CPU</i> 30 N Gould St suite 4000 Sheridan WY 82801		
Size: A4	Number:*	Revision: 0.1			
Date: 11/26/2020	Time: 4:04:18 PM	Sheet* of *			
File: <u>TPS61023_Boost_Converter.SchDoc</u>					

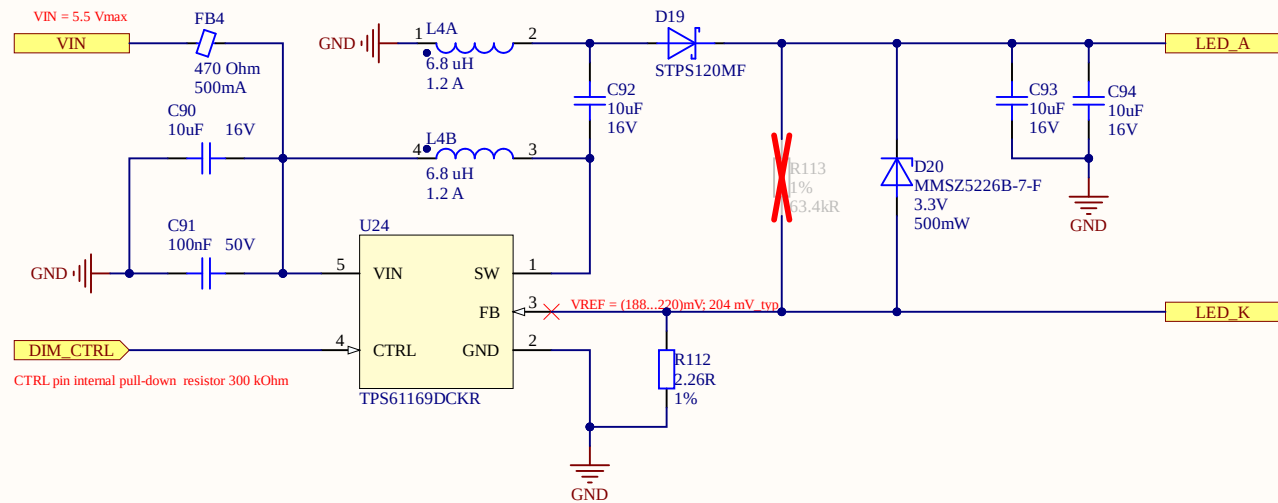


Title +3V3 Output DC-DC Converter			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number: *	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:18 PM	Sheet * of *		
File: TPS6209733_Buck_2A_3V3_FixedOutput.SchDoc				

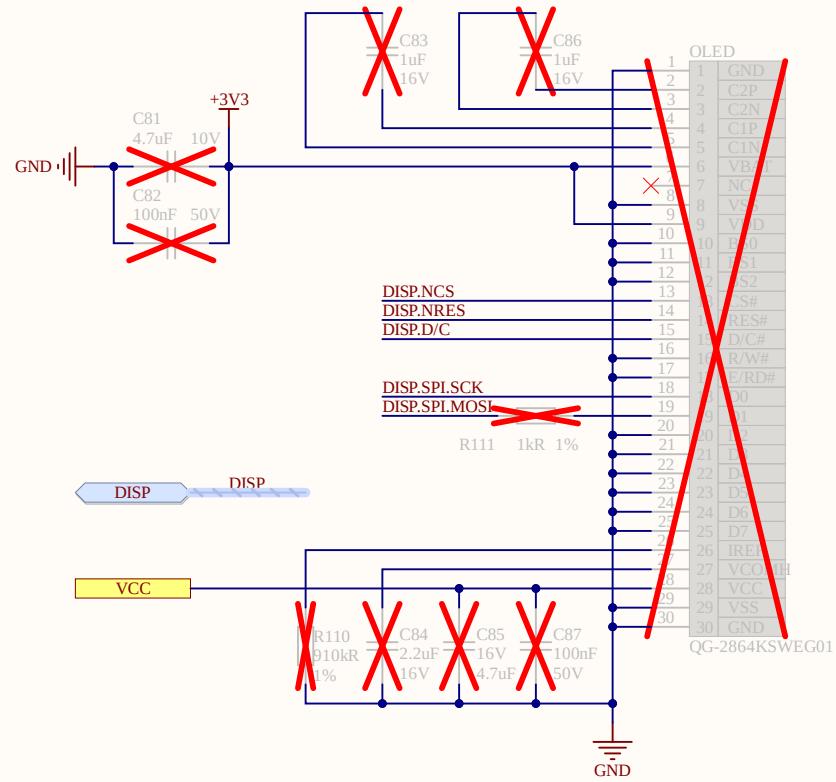


Title LDLN025 +1V8 Fix Voltage Regulator			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number:*	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:18 PM	Sheet* of *		
File: LDLN025_1V8_Regulator.SchDoc				

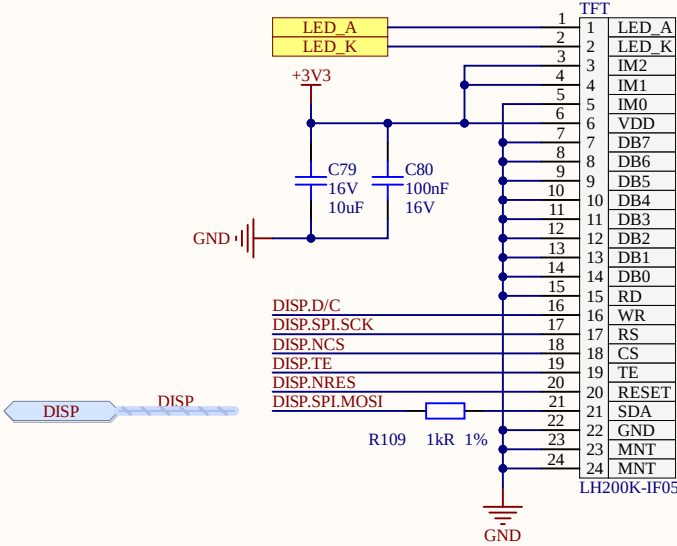




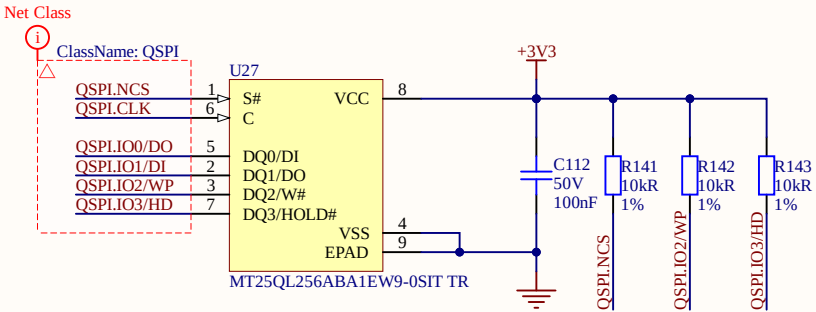
Title TPS61169 SEPIC 100 mA LED Driver			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number: *	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:18 PM	Sheet * of *		
File: TPS61169_SEPIC_90_mA_LED_Driver.SchDoc				



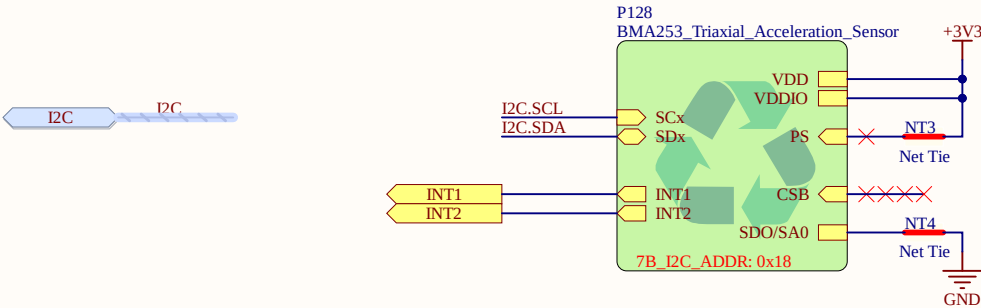
Title QG-2864KSWEG01 Display Module			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number: *	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:18 PM	Sheet* of *		
File: QG-2864KSWEG01_OLED_Display_Module.SchDoc				



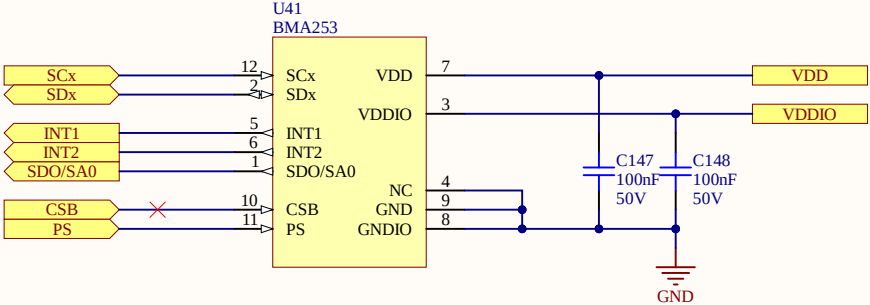
Title LH200K-IF05 2" 240x320 TFT LCD			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number:*	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:18 PM	Sheet* of *		
File: LH200K-IF05_240x320_TFT_LCD_Module.SchDoc				



Title Quad SPI Flash Memory			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number:*	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:19 PM	Sheet* of *		
File: MT25QL256_QSPI_NOR_Flash.SchDoc				



Title BMA253 Orientation Sensor			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number:20	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:19 PM	Sheet20 of 50		
File: BMA253_Orientation_Sensor.SchDoc				



Title BMA253 Digital Accelerometer			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number:*	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:19 PM	Sheet* of *		
File: BMA253_Triaxial_Acceleration_Sensor.SchDoc				

A

B

C

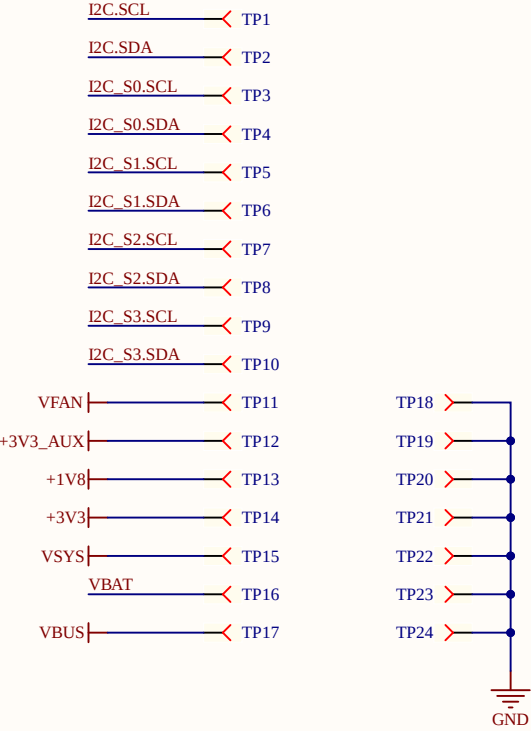
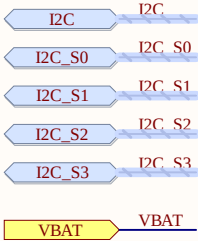
D

A

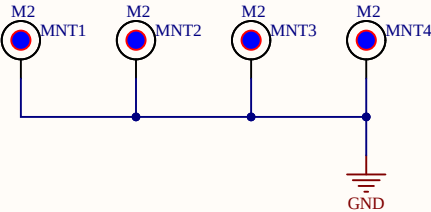
B

C

D

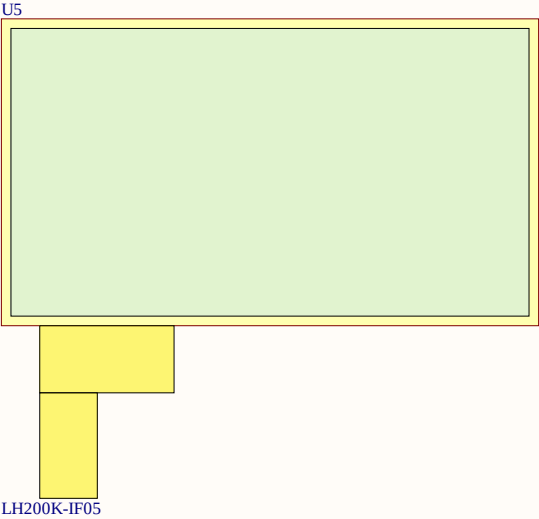


Title Test Points			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number:21	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:19 PM	Sheet21 of 50		
File: Test_Points.SchDoc				



G1
ESD
Logo_ESD

Logo1
Acme CPU Logo
Logo ACME_CPU



Title Auxiliary Components			Acme_CPU 30 N Gould St suite 4000 Sheridan WY 82801	
Size: A4	Number:22	Revision: 0.1		
Date: 11/26/2020	Time: 4:04:19 PM	Sheet22 of 50		
File: Auxiliary.SchDoc				